Tiny Engineer - laptop screen inserts: Colour screens

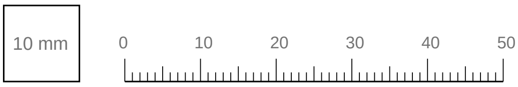
Servo preset: Tower Pro SG90 (sg90)

Print only one page for sg90 parts.

Colour or black-only printer on white paper. Dark fills hide the paper colour.

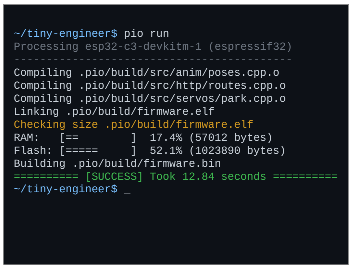
Insert 45.6 x 34.4 mm (cut on the grey line). Visible window 45.2 x 25.9 mm, 0.2 mm from the left and 1.7 mm from the top edge.

Slide the paper insert, then a same-size piece of clear PET, into the lid pocket in front of LaptopScreen.



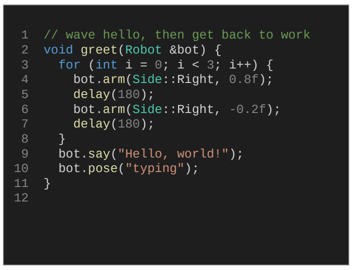
Square must measure 10 x 10 mm and the ruler 50 mm.

If not, reprint at 100% / Actual size (scaling off).



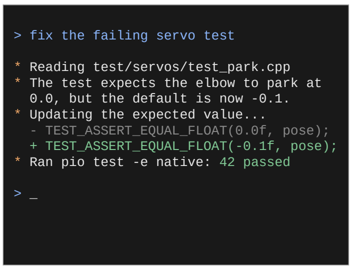
```
~/tiny-engineer$ pio run
Processing esp32-c3-devkitm-1 (espressif32)
-------------------------------------------
Compiling .pio/build/src/anim/poses.cpp.o
Compiling .pio/build/src/http/routes.cpp.o
Compiling .pio/build/src/servos/park.cpp.o
Linking .pio/build/firmware.elf
Checking size .pio/build/firmware.elf
RAM:   [==        ]  17.4% (57012 bytes)
Flash: [=====     ]  52.1% (1023890 bytes)
Building .pio/build/firmware.bin
========== [SUCCESS] Took 12.84 seconds ==========
~/tiny-engineer$ _
```

sg90 - Build output



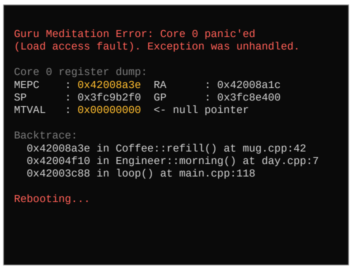
```
// wave hello, then get back to work
void greet(Robot &bot) {
  for (int i = 0; i < 3; i++) {
    bot.arm(Side::Right, 0.8f);
    delay(180);
    bot.arm(Side::Right, -0.2f);
    delay(180);
  }
  bot.say("Hello, world!");
  bot.pose("typing");
}
```

sg90 - Source code



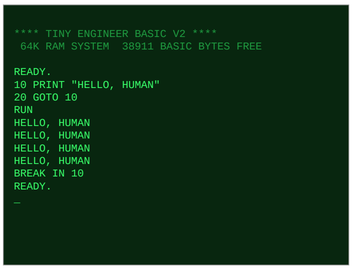
```
> fix the failing servo test

* Reading test/servos/test_park.cpp
* The test expects the elbow to park at
  0.0, but the default is now -0.1.
* Updating the expected value...
  - TEST_ASSERT_EQUAL_FLOAT(0.0f, pose);
  + TEST_ASSERT_EQUAL_FLOAT(-0.1f, pose);
* Ran pio test -e native: 42 passed

> _
```

sg90 - AI agent chat



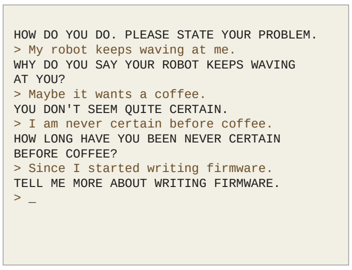
```
Guru Meditation Error: Core 0 panic'ed
(Load access fault). Exception was unhandled.

Core 0 register dump:
MEPC    : 0x42008a3e  RA      : 0x42008a1c
SP      : 0x3fc9b2f0  GP      : 0x3fc8e400
MTVAL   : 0x00000000  <- null pointer

Backtrace:
  0x42008a3e in Coffee::refill() at mug.cpp:42
  0x42004f10 in Engineer::morning() at day.cpp:7
  0x42003c88 in loop() at main.cpp:118

Rebooting...
```

sg90 - Crash / stack trace



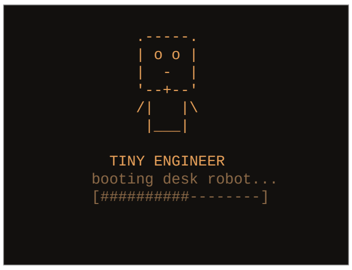
```
**** TINY ENGINEER BASIC V2 ****
 64K RAM SYSTEM  38911 BASIC BYTES FREE

READY.
10 PRINT "HELLO, HUMAN"
20 GOTO 10
RUN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
BREAK IN 10
READY.
_
```

sg90 - Retro computer



```
HOW DO YOU DO. PLEASE STATE YOUR PROBLEM.
> My robot keeps waving at me.
WHY DO YOU SAY YOUR ROBOT KEEPS WAVING
AT YOU?
> Maybe it wants a coffee.
YOU DON'T SEEM QUITE CERTAIN.
> I am never certain before coffee.
HOW LONG HAVE YOU BEEN NEVER CERTAIN
BEFORE COFFEE?
> Since I started writing firmware.
TELL ME MORE ABOUT WRITING FIRMWARE.
> _
```

sg90 - ELIZA-style chat

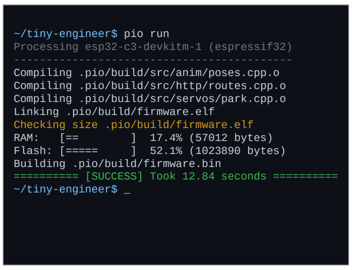


```
   .-----.
   | o o |
   |  -  |
   '--+--'
  /|     |\
    |___|

 TINY ENGINEER
booting desk robot...
[##########--------]
```

sg90 - Tiny Engineer splash

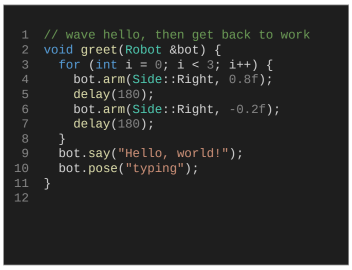
draw or glue your screen inside the dashed line (visible area)

sg90 - Blank template



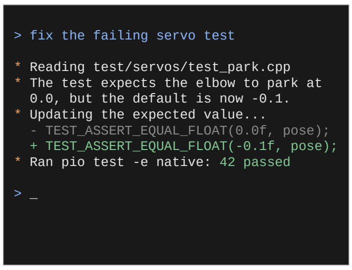
```
~/tiny-engineer$ pio run
Processing esp32-c3-devkitm-1 (espressif32)
-------------------------------------------
Compiling .pio/build/src/anim/poses.cpp.o
Compiling .pio/build/src/http/routes.cpp.o
Compiling .pio/build/src/servos/park.cpp.o
Linking .pio/build/firmware.elf
Checking size .pio/build/firmware.elf
RAM:   [==        ]  17.4% (57012 bytes)
Flash: [=====     ]  52.1% (1023890 bytes)
Building .pio/build/firmware.bin
========== [SUCCESS] Took 12.84 seconds ==========
~/tiny-engineer$ _
```

sg90 - Build output



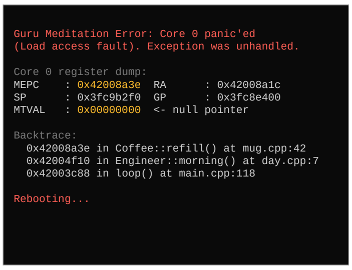
```
// wave hello, then get back to work
void greet(Robot &bot) {
  for (int i = 0; i < 3; i++) {
    bot.arm(Side::Right, 0.8f);
    delay(180);
    bot.arm(Side::Right, -0.2f);
    delay(180);
  }
  bot.say("Hello, world!");
  bot.pose("typing");
}
```

sg90 - Source code



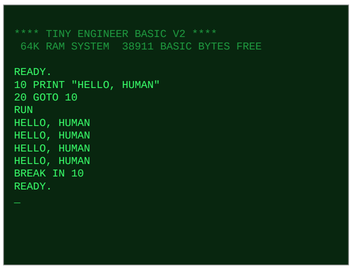
```
> fix the failing servo test

* Reading test/servos/test_park.cpp
* The test expects the elbow to park at
  0.0, but the default is now -0.1.
* Updating the expected value...
  - TEST_ASSERT_EQUAL_FLOAT(0.0f, pose);
  + TEST_ASSERT_EQUAL_FLOAT(-0.1f, pose);
* Ran pio test -e native: 42 passed

> _
```

sg90 - AI agent chat



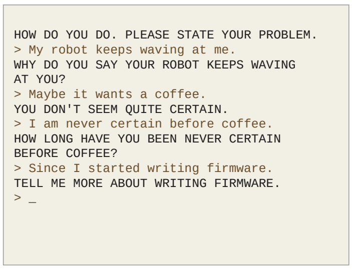
```
Guru Meditation Error: Core 0 panic'ed
(Load access fault). Exception was unhandled.

Core 0 register dump:
MEPC    : 0x42008a3e  RA      : 0x42008a1c
SP      : 0x3fc9b2f0  GP      : 0x3fc8e400
MTVAL   : 0x00000000  <- null pointer

Backtrace:
  0x42008a3e in Coffee::refill() at mug.cpp:42
  0x42004f10 in Engineer::morning() at day.cpp:7
  0x42003c88 in loop() at main.cpp:118

Rebooting...
```

sg90 - Crash / stack trace



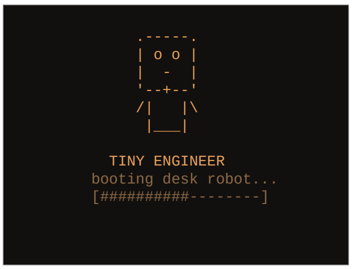
```
**** TINY ENGINEER BASIC V2 ****
 64K RAM SYSTEM  38911 BASIC BYTES FREE

READY.
10 PRINT "HELLO, HUMAN"
20 GOTO 10
RUN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
BREAK IN 10
READY.
_
```

sg90 - Retro computer



```
HOW DO YOU DO. PLEASE STATE YOUR PROBLEM.
> My robot keeps waving at me.
WHY DO YOU SAY YOUR ROBOT KEEPS WAVING
AT YOU?
> Maybe it wants a coffee.
YOU DON'T SEEM QUITE CERTAIN.
> I am never certain before coffee.
HOW LONG HAVE YOU BEEN NEVER CERTAIN
BEFORE COFFEE?
> Since I started writing firmware.
TELL ME MORE ABOUT WRITING FIRMWARE.
> _
```

sg90 - ELIZA-style chat



```
   .-----.
   | o o |
   |  -  |
   '--+--'
  /|     |\
    |___|

 TINY ENGINEER
booting desk robot...
[##########--------]
```

sg90 - Tiny Engineer splash

Tiny Engineer - laptop screen inserts: Paper screens

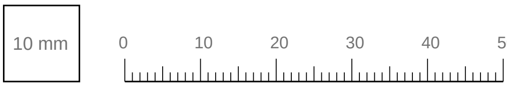
Servo preset: Tower Pro SG90 (sg90)

Print only one page for sg90 parts.

Any printer, any paper colour: no fill, so the paper becomes the screen.

Insert 45.6 x 34.4 mm (cut on the grey line). Visible window 45.2 x 25.9 mm, 0.2 mm from the left and 1.7 mm from the top edge.

Slide the paper insert, then a same-size piece of clear PET, into the lid pocket in front of LaptopScreen.



Square must measure 10 x 10 mm and the ruler 50 mm.

If not, reprint at 100% / Actual size (scaling off).

```
~/tiny-engineer$ pio run
Processing esp32-c3-devkitm-1 (espressif32)
------------------------------------------
Compiling .pio/build/src/anim/poses.cpp.o
Compiling .pio/build/src/http/routes.cpp.o
Compiling .pio/build/src/servos/park.cpp.o
Linking .pio/build/firmware.elf
Checking size .pio/build/firmware.elf
RAM:   [==        ]  17.4% (57012 bytes)
Flash: [=====     ]  52.1% (1023890 bytes)
Building .pio/build/firmware.bin
========== [SUCCESS] Took 12.84 seconds ==========
~/tiny-engineer$ _
```

sg90 - Build output

```
// wave hello, then get back to work
void greet(Robot &bot) {
  for (int i = 0; i < 3; i++) {
    bot.arm(Side::Right, 0.8f);
    delay(180);
    bot.arm(Side::Right, -0.2f);
    delay(180);
  }
  bot.say("Hello, world!");
  bot.pose("typing");
}
```

sg90 - Source code

```
> fix the failing servo test

* Reading test/servos/test_park.cpp
* The test expects the elbow to park at
  0.0, but the default is now -0.1.
* Updating the expected value...
  - TEST_ASSERT_EQUAL_FLOAT(0.0f, pose);
  + TEST_ASSERT_EQUAL_FLOAT(-0.1f, pose);
* Ran pio test -e native: 42 passed

> _
```

sg90 - AI agent chat

```
Guru Meditation Error: Core 0 panic'ed
(Load access fault). Exception was unhandled.

Core 0 register dump:
MEPC    : 0x42008a3e  RA      : 0x42008a1c
SP      : 0x3fc9b2f0  GP      : 0x3fc8e400
MTVAL   : 0x00000000  <- null pointer

Backtrace:
  0x42008a3e in Coffee::refill() at mug.cpp:42
  0x42004f10 in Engineer::morning() at day.cpp:7
  0x42003c88 in loop() at main.cpp:118

Rebooting...
```

sg90 - Crash / stack trace

```
**** TINY ENGINEER BASIC V2 ****
 64K RAM SYSTEM  38911 BASIC BYTES FREE

READY.
10 PRINT "HELLO, HUMAN"
20 GOTO 10
RUN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
BREAK IN 10
READY.
_
```

sg90 - Retro computer

```
HOW DO YOU DO. PLEASE STATE YOUR PROBLEM.
> My robot keeps waving at me.
WHY DO YOU SAY YOUR ROBOT KEEPS WAVING
AT YOU?
> Maybe it wants a coffee.
YOU DON'T SEEM QUITE CERTAIN.
> I am never certain before coffee.
HOW LONG HAVE YOU BEEN NEVER CERTAIN
BEFORE COFFEE?
> Since I started writing firmware.
TELL ME MORE ABOUT WRITING FIRMWARE.
> _
```

sg90 - ELIZA-style chat

```
        .-----.
        | o o |
        |  -  |
        '--+--'
        /|   |\
         |___|

      TINY ENGINEER
   booting desk robot...
   [##########--------]
```

sg90 - Tiny Engineer splash

```
draw or glue your screen inside
the dashed line (visible area)
```

sg90 - Blank template

```
~/tiny-engineer$ pio run
Processing esp32-c3-devkitm-1 (espressif32)
------------------------------------------
Compiling .pio/build/src/anim/poses.cpp.o
Compiling .pio/build/src/http/routes.cpp.o
Compiling .pio/build/src/servos/park.cpp.o
Linking .pio/build/firmware.elf
Checking size .pio/build/firmware.elf
RAM:   [==        ]  17.4% (57012 bytes)
Flash: [=====     ]  52.1% (1023890 bytes)
Building .pio/build/firmware.bin
========== [SUCCESS] Took 12.84 seconds ==========
~/tiny-engineer$ _
```

sg90 - Build output

```
// wave hello, then get back to work
void greet(Robot &bot) {
  for (int i = 0; i < 3; i++) {
    bot.arm(Side::Right, 0.8f);
    delay(180);
    bot.arm(Side::Right, -0.2f);
    delay(180);
  }
  bot.say("Hello, world!");
  bot.pose("typing");
}
```

sg90 - Source code

```
> fix the failing servo test

* Reading test/servos/test_park.cpp
* The test expects the elbow to park at
  0.0, but the default is now -0.1.
* Updating the expected value...
  - TEST_ASSERT_EQUAL_FLOAT(0.0f, pose);
  + TEST_ASSERT_EQUAL_FLOAT(-0.1f, pose);
* Ran pio test -e native: 42 passed

> _
```

sg90 - AI agent chat

```
Guru Meditation Error: Core 0 panic'ed
(Load access fault). Exception was unhandled.

Core 0 register dump:
MEPC    : 0x42008a3e  RA      : 0x42008a1c
SP      : 0x3fc9b2f0  GP      : 0x3fc8e400
MTVAL   : 0x00000000  <- null pointer

Backtrace:
  0x42008a3e in Coffee::refill() at mug.cpp:42
  0x42004f10 in Engineer::morning() at day.cpp:7
  0x42003c88 in loop() at main.cpp:118

Rebooting...
```

sg90 - Crash / stack trace

```
**** TINY ENGINEER BASIC V2 ****
 64K RAM SYSTEM  38911 BASIC BYTES FREE

READY.
10 PRINT "HELLO, HUMAN"
20 GOTO 10
RUN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
BREAK IN 10
READY.
_
```

sg90 - Retro computer

```
HOW DO YOU DO. PLEASE STATE YOUR PROBLEM.
> My robot keeps waving at me.
WHY DO YOU SAY YOUR ROBOT KEEPS WAVING
AT YOU?
> Maybe it wants a coffee.
YOU DON'T SEEM QUITE CERTAIN.
> I am never certain before coffee.
HOW LONG HAVE YOU BEEN NEVER CERTAIN
BEFORE COFFEE?
> Since I started writing firmware.
TELL ME MORE ABOUT WRITING FIRMWARE.
> _
```

sg90 - ELIZA-style chat

```
        .-----.
        | o o |
        |  -  |
        '--+--'
        /|   |\
         |___|

      TINY ENGINEER
   booting desk robot...
   [##########--------]
```

sg90 - Tiny Engineer splash

Tiny Engineer - laptop screen inserts: Colour screens

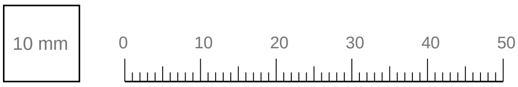
Servo preset: Feetech FS0307 (fs0307)

Print only one page for fs0307 parts.

Colour or black-only printer on white paper. Dark fills hide the paper colour.

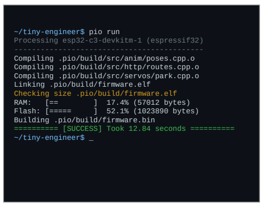
Insert 33.8 x 26.0 mm (cut on the grey line). Visible window 33.4 x 17.5 mm, 0.2 mm from the left and 1.7 mm from the top edge.

Slide the paper insert, then a same-size piece of clear PET, into the lid pocket in front of LaptopScreen.



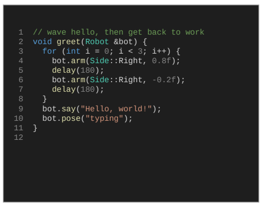
Square must measure 10 x 10 mm and the ruler 50 mm.

If not, reprint at 100% / Actual size (scaling off).



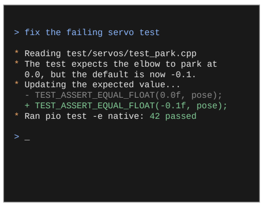
```
~/tiny-engineer$ pio run
Processing esp32-c3-devkitm-1 (espressif32)
-------------------------------------------
Compiling .pio/build/src/anim/poses.cpp.o
Compiling .pio/build/src/http/routes.cpp.o
Compiling .pio/build/src/servos/park.cpp.o
Linking .pio/build/firmware.elf
Checking size .pio/build/firmware.elf
RAM:   [==        ]  17.4% (57012 bytes)
Flash: [=====     ]  52.1% (1023890 bytes)
Building .pio/build/firmware.bin
========== [SUCCESS] Took 12.84 seconds ==========
~/tiny-engineer$ _
```

fs0307 - Build output



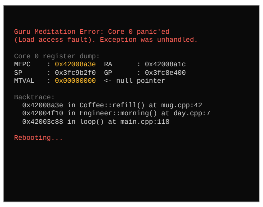
```
// wave hello, then get back to work
void greet(Robot &bot) {
  for (int i = 0; i < 3; i++) {
    bot.arm(Side::Right, 0.8f);
    delay(180);
    bot.arm(Side::Right, -0.2f);
    delay(180);
  }
  bot.say("Hello, world!");
  bot.pose("typing");
}
```

fs0307 - Source code



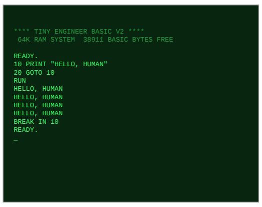
```
> fix the failing servo test

* Reading test/servos/test_park.cpp
* The test expects the elbow to park at
  0.0, but the default is now -0.1.
* Updating the expected value...
  - TEST_ASSERT_EQUAL_FLOAT(0.0f, pose);
  + TEST_ASSERT_EQUAL_FLOAT(-0.1f, pose);
* Ran pio test -e native: 42 passed

> _
```

fs0307 - AI agent chat



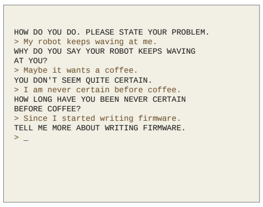
```
Guru Meditation Error: Core 0 panic'ed
(Load access fault). Exception was unhandled.

Core 0 register dump:
MEPC    : 0x42008a3e  RA      : 0x42008a1c
SP      : 0x3fc9b2f0  GP      : 0x3fc8e400
MTVAL   : 0x00000000  <- null pointer

Backtrace:
  0x42008a3e in Coffee::refill() at mug.cpp:42
  0x42004f10 in Engineer::morning() at day.cpp:7
  0x42003c88 in loop() at main.cpp:118

Rebooting...
```

fs0307 - Crash / stack trace



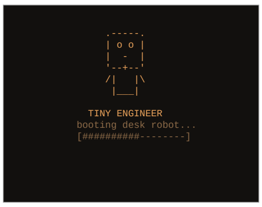
```
**** TINY ENGINEER BASIC V2 ****
 64K RAM SYSTEM  38911 BASIC BYTES FREE

READY.
10 PRINT "HELLO, HUMAN"
20 GOTO 10
RUN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
BREAK IN 10
READY.
_
```

fs0307 - Retro computer



```
HOW DO YOU DO. PLEASE STATE YOUR PROBLEM.
> My robot keeps waving at me.
WHY DO YOU SAY YOUR ROBOT KEEPS WAVING
AT YOU?
> Maybe it wants a coffee.
YOU DON'T SEEM QUITE CERTAIN.
> I am never certain before coffee.
HOW LONG HAVE YOU BEEN NEVER CERTAIN
BEFORE COFFEE?
> Since I started writing firmware.
TELL ME MORE ABOUT WRITING FIRMWARE.
> _
```

fs0307 - ELIZA-style chat



```
     .-----.
     | o o |
     |  -  |
     '--+--'
    /|     |\
     |_____|

   TINY ENGINEER
 booting desk robot...
[##########--------]
```

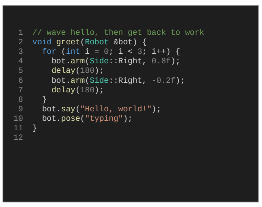
fs0307 - Tiny Engineer splash

draw or glue your screen inside
the dashed line (visible area)

fs0307 - Blank template

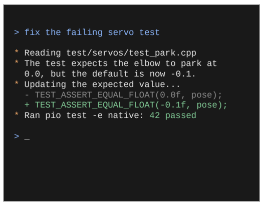
```
~/tiny-engineer$ pio run
Processing esp32-c3-devkitm-1 (espressif32)
-------------------------------------------
Compiling .pio/build/src/anim/poses.cpp.o
Compiling .pio/build/src/http/routes.cpp.o
Compiling .pio/build/src/servos/park.cpp.o
Linking .pio/build/firmware.elf
Checking size .pio/build/firmware.elf
RAM:   [==        ]  17.4% (57012 bytes)
Flash: [=====     ]  52.1% (1023890 bytes)
Building .pio/build/firmware.bin
========== [SUCCESS] Took 12.84 seconds ==========
~/tiny-engineer$ _
```

fs0307 - Build output



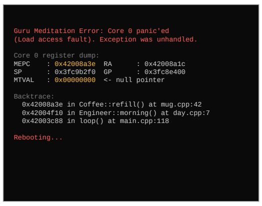
```
// wave hello, then get back to work
void greet(Robot &bot) {
  for (int i = 0; i < 3; i++) {
    bot.arm(Side::Right, 0.8f);
    delay(180);
    bot.arm(Side::Right, -0.2f);
    delay(180);
  }
  bot.say("Hello, world!");
  bot.pose("typing");
}
```

fs0307 - Source code



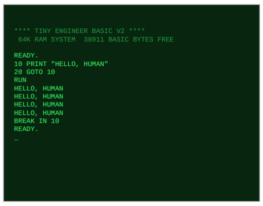
```
> fix the failing servo test

* Reading test/servos/test_park.cpp
* The test expects the elbow to park at
  0.0, but the default is now -0.1.
* Updating the expected value...
  - TEST_ASSERT_EQUAL_FLOAT(0.0f, pose);
  + TEST_ASSERT_EQUAL_FLOAT(-0.1f, pose);
* Ran pio test -e native: 42 passed

> _
```

fs0307 - AI agent chat



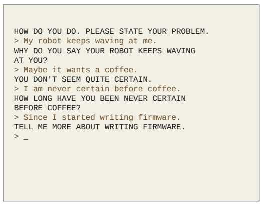
```
Guru Meditation Error: Core 0 panic'ed
(Load access fault). Exception was unhandled.

Core 0 register dump:
MEPC    : 0x42008a3e  RA      : 0x42008a1c
SP      : 0x3fc9b2f0  GP      : 0x3fc8e400
MTVAL   : 0x00000000  <- null pointer

Backtrace:
  0x42008a3e in Coffee::refill() at mug.cpp:42
  0x42004f10 in Engineer::morning() at day.cpp:7
  0x42003c88 in loop() at main.cpp:118

Rebooting...
```

fs0307 - Crash / stack trace



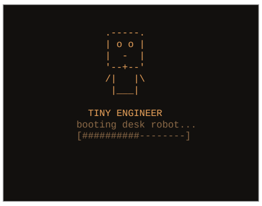
```
**** TINY ENGINEER BASIC V2 ****
 64K RAM SYSTEM  38911 BASIC BYTES FREE

READY.
10 PRINT "HELLO, HUMAN"
20 GOTO 10
RUN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
BREAK IN 10
READY.
_
```

fs0307 - Retro computer



```
HOW DO YOU DO. PLEASE STATE YOUR PROBLEM.
> My robot keeps waving at me.
WHY DO YOU SAY YOUR ROBOT KEEPS WAVING
AT YOU?
> Maybe it wants a coffee.
YOU DON'T SEEM QUITE CERTAIN.
> I am never certain before coffee.
HOW LONG HAVE YOU BEEN NEVER CERTAIN
BEFORE COFFEE?
> Since I started writing firmware.
TELL ME MORE ABOUT WRITING FIRMWARE.
> _
```

fs0307 - ELIZA-style chat

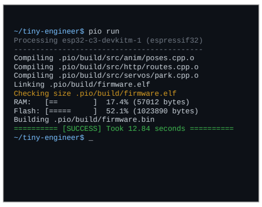


```
     .-----.
     | o o |
     |  -  |
     '--+--'
    /|     |\
     |_____|

   TINY ENGINEER
 booting desk robot...
[##########--------]
```

fs0307 - Tiny Engineer splash

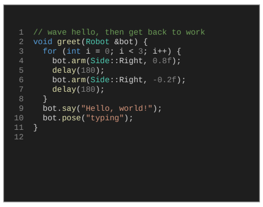
draw or glue your screen inside
the dashed line (visible area)

fs0307 - Blank template



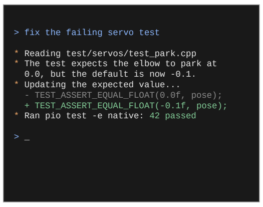
```
~/tiny-engineer$ pio run
Processing esp32-c3-devkitm-1 (espressif32)
-------------------------------------------
Compiling .pio/build/src/anim/poses.cpp.o
Compiling .pio/build/src/http/routes.cpp.o
Compiling .pio/build/src/servos/park.cpp.o
Linking .pio/build/firmware.elf
Checking size .pio/build/firmware.elf
RAM:   [==        ]  17.4% (57012 bytes)
Flash: [=====     ]  52.1% (1023890 bytes)
Building .pio/build/firmware.bin
========== [SUCCESS] Took 12.84 seconds ==========
~/tiny-engineer$ _
```

fs0307 - Build output



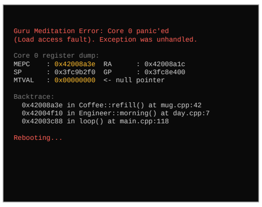
```
// wave hello, then get back to work
void greet(Robot &bot) {
  for (int i = 0; i < 3; i++) {
    bot.arm(Side::Right, 0.8f);
    delay(180);
    bot.arm(Side::Right, -0.2f);
    delay(180);
  }
  bot.say("Hello, world!");
  bot.pose("typing");
}
```

fs0307 - Source code



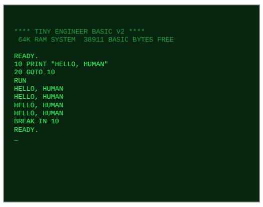
```
> fix the failing servo test

* Reading test/servos/test_park.cpp
* The test expects the elbow to park at
  0.0, but the default is now -0.1.
* Updating the expected value...
  - TEST_ASSERT_EQUAL_FLOAT(0.0f, pose);
  + TEST_ASSERT_EQUAL_FLOAT(-0.1f, pose);
* Ran pio test -e native: 42 passed

> _
```

fs0307 - AI agent chat



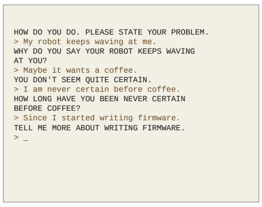
```
Guru Meditation Error: Core 0 panic'ed
(Load access fault). Exception was unhandled.

Core 0 register dump:
MEPC    : 0x42008a3e  RA      : 0x42008a1c
SP      : 0x3fc9b2f0  GP      : 0x3fc8e400
MTVAL   : 0x00000000  <- null pointer

Backtrace:
  0x42008a3e in Coffee::refill() at mug.cpp:42
  0x42004f10 in Engineer::morning() at day.cpp:7
  0x42003c88 in loop() at main.cpp:118

Rebooting...
```

fs0307 - Crash / stack trace



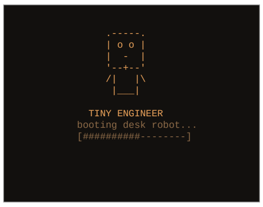
```
**** TINY ENGINEER BASIC V2 ****
 64K RAM SYSTEM  38911 BASIC BYTES FREE

READY.
10 PRINT "HELLO, HUMAN"
20 GOTO 10
RUN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
BREAK IN 10
READY.
_
```

fs0307 - Retro computer



```
HOW DO YOU DO. PLEASE STATE YOUR PROBLEM.
> My robot keeps waving at me.
WHY DO YOU SAY YOUR ROBOT KEEPS WAVING
AT YOU?
> Maybe it wants a coffee.
YOU DON'T SEEM QUITE CERTAIN.
> I am never certain before coffee.
HOW LONG HAVE YOU BEEN NEVER CERTAIN
BEFORE COFFEE?
> Since I started writing firmware.
TELL ME MORE ABOUT WRITING FIRMWARE.
> _
```

fs0307 - ELIZA-style chat



```
     .-----.
     | o o |
     |  -  |
     '--+--'
    /|     |\
     |_____|

   TINY ENGINEER
 booting desk robot...
[##########--------]
```

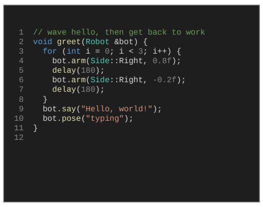
fs0307 - Tiny Engineer splash

draw or glue your screen inside
the dashed line (visible area)

fs0307 - Blank template

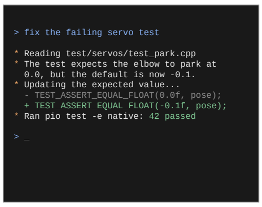
```
~/tiny-engineer$ pio run
Processing esp32-c3-devkitm-1 (espressif32)
-------------------------------------------
Compiling .pio/build/src/anim/poses.cpp.o
Compiling .pio/build/src/http/routes.cpp.o
Compiling .pio/build/src/servos/park.cpp.o
Linking .pio/build/firmware.elf
Checking size .pio/build/firmware.elf
RAM:   [==        ]  17.4% (57012 bytes)
Flash: [=====     ]  52.1% (1023890 bytes)
Building .pio/build/firmware.bin
========== [SUCCESS] Took 12.84 seconds ==========
~/tiny-engineer$ _
```

fs0307 - Build output



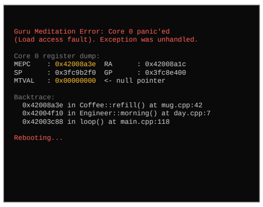
```
// wave hello, then get back to work
void greet(Robot &bot) {
  for (int i = 0; i < 3; i++) {
    bot.arm(Side::Right, 0.8f);
    delay(180);
    bot.arm(Side::Right, -0.2f);
    delay(180);
  }
  bot.say("Hello, world!");
  bot.pose("typing");
}
```

fs0307 - Source code



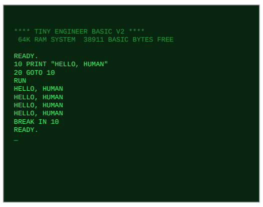
```
> fix the failing servo test

* Reading test/servos/test_park.cpp
* The test expects the elbow to park at
  0.0, but the default is now -0.1.
* Updating the expected value...
  - TEST_ASSERT_EQUAL_FLOAT(0.0f, pose);
  + TEST_ASSERT_EQUAL_FLOAT(-0.1f, pose);
* Ran pio test -e native: 42 passed

> _
```

fs0307 - AI agent chat



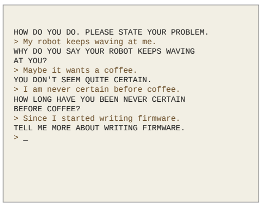
```
Guru Meditation Error: Core 0 panic'ed
(Load access fault). Exception was unhandled.

Core 0 register dump:
MEPC    : 0x42008a3e  RA      : 0x42008a1c
SP      : 0x3fc9b2f0  GP      : 0x3fc8e400
MTVAL   : 0x00000000  <- null pointer

Backtrace:
  0x42008a3e in Coffee::refill() at mug.cpp:42
  0x42004f10 in Engineer::morning() at day.cpp:7
  0x42003c88 in loop() at main.cpp:118

Rebooting...
```

fs0307 - Crash / stack trace



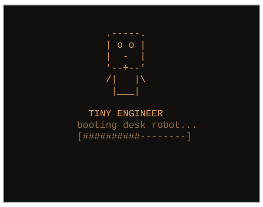
```
**** TINY ENGINEER BASIC V2 ****
 64K RAM SYSTEM  38911 BASIC BYTES FREE

READY.
10 PRINT "HELLO, HUMAN"
20 GOTO 10
RUN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
BREAK IN 10
READY.
_
```

fs0307 - Retro computer



```
HOW DO YOU DO. PLEASE STATE YOUR PROBLEM.
> My robot keeps waving at me.
WHY DO YOU SAY YOUR ROBOT KEEPS WAVING
AT YOU?
> Maybe it wants a coffee.
YOU DON'T SEEM QUITE CERTAIN.
> I am never certain before coffee.
HOW LONG HAVE YOU BEEN NEVER CERTAIN
BEFORE COFFEE?
> Since I started writing firmware.
TELL ME MORE ABOUT WRITING FIRMWARE.
> _
```

fs0307 - ELIZA-style chat

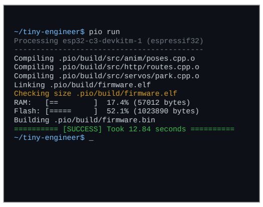


```
     .-----.
     | o o |
     |  -  |
     '--+--'
    /|     |\
     |_____|

   TINY ENGINEER
 booting desk robot...
[##########--------]
```

fs0307 - Tiny Engineer splash

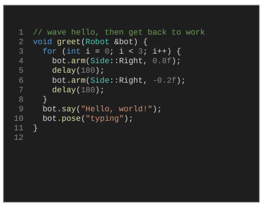
draw or glue your screen inside
the dashed line (visible area)

fs0307 - Blank template



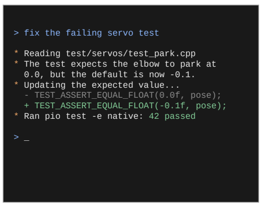
```
~/tiny-engineer$ pio run
Processing esp32-c3-devkitm-1 (espressif32)
-------------------------------------------
Compiling .pio/build/src/anim/poses.cpp.o
Compiling .pio/build/src/http/routes.cpp.o
Compiling .pio/build/src/servos/park.cpp.o
Linking .pio/build/firmware.elf
Checking size .pio/build/firmware.elf
RAM:   [==        ]  17.4% (57012 bytes)
Flash: [=====     ]  52.1% (1023890 bytes)
Building .pio/build/firmware.bin
========== [SUCCESS] Took 12.84 seconds ==========
~/tiny-engineer$ _
```

fs0307 - Build output



```
// wave hello, then get back to work
void greet(Robot &bot) {
  for (int i = 0; i < 3; i++) {
    bot.arm(Side::Right, 0.8f);
    delay(180);
    bot.arm(Side::Right, -0.2f);
    delay(180);
  }
  bot.say("Hello, world!");
  bot.pose("typing");
}
```

fs0307 - Source code



```
> fix the failing servo test

* Reading test/servos/test_park.cpp
* The test expects the elbow to park at
  0.0, but the default is now -0.1.
* Updating the expected value...
  - TEST_ASSERT_EQUAL_FLOAT(0.0f, pose);
  + TEST_ASSERT_EQUAL_FLOAT(-0.1f, pose);
* Ran pio test -e native: 42 passed

> _
```

fs0307 - AI agent chat

Tiny Engineer - laptop screen inserts: Paper screens

Servo preset: Feetech FS0307 (fs0307)

Print only one page for fs0307 parts.

Any printer, any paper colour: no fill, so the paper becomes the screen.

Insert 33.8 x 26.0 mm (cut on the grey line). Visible window 33.4 x 17.5 mm, 0.2 mm from the left and 1.7 mm from the top edge.

Slide the paper insert, then a same-size piece of clear PET, into the lid pocket in front of LaptopScreen.

10 mm

0 10 20 30 40 50

Square must measure 10 x 10 mm and the ruler 50 mm.

If not, reprint at 100% / Actual size (scaling off).

```
~/tiny-engineer$ pio run
Processing esp32-c3-devkitm-1 (espressif32)
-------------------------------------------
Compiling .pio/build/src/anim/poses.cpp.o
Compiling .pio/build/src/http/routes.cpp.o
Compiling .pio/build/src/servos/park.cpp.o
Linking .pio/build/firmware.elf
Checking size .pio/build/firmware.elf
RAM:   [==        ]  17.4% (57012 bytes)
Flash: [=====     ]  52.1% (1023890 bytes)
Building .pio/build/firmware.bin
========== [SUCCESS] Took 12.84 seconds ==========
~/tiny-engineer$ _
```

fs0307 - Build output

```
// wave hello, then get back to work
void greet(Robot &bot) {
  for (int i = 0; i < 3; i++) {
    bot.arm(Side::Right, 0.8f);
    delay(180);
    bot.arm(Side::Right, -0.2f);
    delay(180);
  }
  bot.say("Hello, world!");
  bot.pose("typing");
}
```

fs0307 - Source code

```
> fix the failing servo test

* Reading test/servos/test_park.cpp
* The test expects the elbow to park at
  0.0, but the default is now -0.1.
* Updating the expected value...
  - TEST_ASSERT_EQUAL_FLOAT(0.0f, pose);
  + TEST_ASSERT_EQUAL_FLOAT(-0.1f, pose);
* Ran pio test -e native: 42 passed

> _
```

fs0307 - AI agent chat

```
Guru Meditation Error: Core 0 panic'ed
(Load access fault). Exception was unhandled.

Core 0 register dump:
MEPC    : 0x42008a3e  RA      : 0x42008a1c
SP      : 0x3fc9b2f0  GP      : 0x3fc8e400
MTVAL   : 0x00000000  <- null pointer

Backtrace:
  0x42008a3e in Coffee::refill() at mug.cpp:42
  0x42004f10 in Engineer::morning() at day.cpp:7
  0x42003c88 in loop() at main.cpp:118

Rebooting...
```

fs0307 - Crash / stack trace

```
**** TINY ENGINEER BASIC V2 ****
 64K RAM SYSTEM  38911 BASIC BYTES FREE

READY.
10 PRINT "HELLO, HUMAN"
20 GOTO 10
RUN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
BREAK IN 10
READY.
_
```

fs0307 - Retro computer

```
HOW DO YOU DO. PLEASE STATE YOUR PROBLEM.
> My robot keeps waving at me.
WHY DO YOU SAY YOUR ROBOT KEEPS WAVING
AT YOU?
> Maybe it wants a coffee.
YOU DON'T SEEM QUITE CERTAIN.
> I am never certain before coffee.
HOW LONG HAVE YOU BEEN NEVER CERTAIN
BEFORE COFFEE?
> Since I started writing firmware.
TELL ME MORE ABOUT WRITING FIRMWARE.
> _
```

fs0307 - ELIZA-style chat

```
       .-----.
       | o o |
       |  -  |
       '--+--'
      /|     |\
       |_____|

     TINY ENGINEER
   booting desk robot...
   [##########--------]
```

fs0307 - Tiny Engineer splash

```
draw or glue your screen inside
the dashed line (visible area)
```

fs0307 - Blank template

```
~/tiny-engineer$ pio run
Processing esp32-c3-devkitm-1 (espressif32)
-------------------------------------------
Compiling .pio/build/src/anim/poses.cpp.o
Compiling .pio/build/src/http/routes.cpp.o
Compiling .pio/build/src/servos/park.cpp.o
Linking .pio/build/firmware.elf
Checking size .pio/build/firmware.elf
RAM:   [==        ]  17.4% (57012 bytes)
Flash: [=====     ]  52.1% (1023890 bytes)
Building .pio/build/firmware.bin
========== [SUCCESS] Took 12.84 seconds ==========
~/tiny-engineer$ _
```

fs0307 - Build output

```
// wave hello, then get back to work
void greet(Robot &bot) {
  for (int i = 0; i < 3; i++) {
    bot.arm(Side::Right, 0.8f);
    delay(180);
    bot.arm(Side::Right, -0.2f);
    delay(180);
  }
  bot.say("Hello, world!");
  bot.pose("typing");
}
```

fs0307 - Source code

```
> fix the failing servo test

* Reading test/servos/test_park.cpp
* The test expects the elbow to park at
  0.0, but the default is now -0.1.
* Updating the expected value...
  - TEST_ASSERT_EQUAL_FLOAT(0.0f, pose);
  + TEST_ASSERT_EQUAL_FLOAT(-0.1f, pose);
* Ran pio test -e native: 42 passed

> _
```

fs0307 - AI agent chat

```
Guru Meditation Error: Core 0 panic'ed
(Load access fault). Exception was unhandled.

Core 0 register dump:
MEPC    : 0x42008a3e  RA      : 0x42008a1c
SP      : 0x3fc9b2f0  GP      : 0x3fc8e400
MTVAL   : 0x00000000  <- null pointer

Backtrace:
  0x42008a3e in Coffee::refill() at mug.cpp:42
  0x42004f10 in Engineer::morning() at day.cpp:7
  0x42003c88 in loop() at main.cpp:118

Rebooting...
```

fs0307 - Crash / stack trace

```
**** TINY ENGINEER BASIC V2 ****
 64K RAM SYSTEM  38911 BASIC BYTES FREE

READY.
10 PRINT "HELLO, HUMAN"
20 GOTO 10
RUN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
BREAK IN 10
READY.
_
```

fs0307 - Retro computer

```
HOW DO YOU DO. PLEASE STATE YOUR PROBLEM.
> My robot keeps waving at me.
WHY DO YOU SAY YOUR ROBOT KEEPS WAVING
AT YOU?
> Maybe it wants a coffee.
YOU DON'T SEEM QUITE CERTAIN.
> I am never certain before coffee.
HOW LONG HAVE YOU BEEN NEVER CERTAIN
BEFORE COFFEE?
> Since I started writing firmware.
TELL ME MORE ABOUT WRITING FIRMWARE.
> _
```

fs0307 - ELIZA-style chat

```
       .-----.
       | o o |
       |  -  |
       '--+--'
      /|     |\
       |_____|

     TINY ENGINEER
   booting desk robot...
   [##########--------]
```

fs0307 - Tiny Engineer splash

```
draw or glue your screen inside
the dashed line (visible area)
```

fs0307 - Blank template

```
~/tiny-engineer$ pio run
Processing esp32-c3-devkitm-1 (espressif32)
-------------------------------------------
Compiling .pio/build/src/anim/poses.cpp.o
Compiling .pio/build/src/http/routes.cpp.o
Compiling .pio/build/src/servos/park.cpp.o
Linking .pio/build/firmware.elf
Checking size .pio/build/firmware.elf
RAM:   [==        ]  17.4% (57012 bytes)
Flash: [=====     ]  52.1% (1023890 bytes)
Building .pio/build/firmware.bin
========== [SUCCESS] Took 12.84 seconds ==========
~/tiny-engineer$ _
```

fs0307 - Build output

```
// wave hello, then get back to work
void greet(Robot &bot) {
  for (int i = 0; i < 3; i++) {
    bot.arm(Side::Right, 0.8f);
    delay(180);
    bot.arm(Side::Right, -0.2f);
    delay(180);
  }
  bot.say("Hello, world!");
  bot.pose("typing");
}
```

fs0307 - Source code

```
> fix the failing servo test

* Reading test/servos/test_park.cpp
* The test expects the elbow to park at
  0.0, but the default is now -0.1.
* Updating the expected value...
  - TEST_ASSERT_EQUAL_FLOAT(0.0f, pose);
  + TEST_ASSERT_EQUAL_FLOAT(-0.1f, pose);
* Ran pio test -e native: 42 passed

> _
```

fs0307 - AI agent chat

```
Guru Meditation Error: Core 0 panic'ed
(Load access fault). Exception was unhandled.

Core 0 register dump:
MEPC    : 0x42008a3e  RA      : 0x42008a1c
SP      : 0x3fc9b2f0  GP      : 0x3fc8e400
MTVAL   : 0x00000000  <- null pointer

Backtrace:
  0x42008a3e in Coffee::refill() at mug.cpp:42
  0x42004f10 in Engineer::morning() at day.cpp:7
  0x42003c88 in loop() at main.cpp:118

Rebooting...
```

fs0307 - Crash / stack trace

```
**** TINY ENGINEER BASIC V2 ****
 64K RAM SYSTEM  38911 BASIC BYTES FREE

READY.
10 PRINT "HELLO, HUMAN"
20 GOTO 10
RUN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
BREAK IN 10
READY.
_
```

fs0307 - Retro computer

```
HOW DO YOU DO. PLEASE STATE YOUR PROBLEM.
> My robot keeps waving at me.
WHY DO YOU SAY YOUR ROBOT KEEPS WAVING
AT YOU?
> Maybe it wants a coffee.
YOU DON'T SEEM QUITE CERTAIN.
> I am never certain before coffee.
HOW LONG HAVE YOU BEEN NEVER CERTAIN
BEFORE COFFEE?
> Since I started writing firmware.
TELL ME MORE ABOUT WRITING FIRMWARE.
> _
```

fs0307 - ELIZA-style chat

```
       .-----.
       | o o |
       |  -  |
       '--+--'
      /|     |\
       |_____|

     TINY ENGINEER
   booting desk robot...
   [##########--------]
```

fs0307 - Tiny Engineer splash

```
draw or glue your screen inside
the dashed line (visible area)
```

fs0307 - Blank template

```
~/tiny-engineer$ pio run
Processing esp32-c3-devkitm-1 (espressif32)
-------------------------------------------
Compiling .pio/build/src/anim/poses.cpp.o
Compiling .pio/build/src/http/routes.cpp.o
Compiling .pio/build/src/servos/park.cpp.o
Linking .pio/build/firmware.elf
Checking size .pio/build/firmware.elf
RAM:   [==        ]  17.4% (57012 bytes)
Flash: [=====     ]  52.1% (1023890 bytes)
Building .pio/build/firmware.bin
========== [SUCCESS] Took 12.84 seconds ==========
~/tiny-engineer$ _
```

fs0307 - Build output

```
// wave hello, then get back to work
void greet(Robot &bot) {
  for (int i = 0; i < 3; i++) {
    bot.arm(Side::Right, 0.8f);
    delay(180);
    bot.arm(Side::Right, -0.2f);
    delay(180);
  }
  bot.say("Hello, world!");
  bot.pose("typing");
}
```

fs0307 - Source code

```
> fix the failing servo test

* Reading test/servos/test_park.cpp
* The test expects the elbow to park at
  0.0, but the default is now -0.1.
* Updating the expected value...
  - TEST_ASSERT_EQUAL_FLOAT(0.0f, pose);
  + TEST_ASSERT_EQUAL_FLOAT(-0.1f, pose);
* Ran pio test -e native: 42 passed

> _
```

fs0307 - AI agent chat

```
Guru Meditation Error: Core 0 panic'ed
(Load access fault). Exception was unhandled.

Core 0 register dump:
MEPC    : 0x42008a3e  RA      : 0x42008a1c
SP      : 0x3fc9b2f0  GP      : 0x3fc8e400
MTVAL   : 0x00000000  <- null pointer

Backtrace:
  0x42008a3e in Coffee::refill() at mug.cpp:42
  0x42004f10 in Engineer::morning() at day.cpp:7
  0x42003c88 in loop() at main.cpp:118

Rebooting...
```

fs0307 - Crash / stack trace

```
**** TINY ENGINEER BASIC V2 ****
 64K RAM SYSTEM  38911 BASIC BYTES FREE

READY.
10 PRINT "HELLO, HUMAN"
20 GOTO 10
RUN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
BREAK IN 10
READY.
_
```

fs0307 - Retro computer

```
HOW DO YOU DO. PLEASE STATE YOUR PROBLEM.
> My robot keeps waving at me.
WHY DO YOU SAY YOUR ROBOT KEEPS WAVING
AT YOU?
> Maybe it wants a coffee.
YOU DON'T SEEM QUITE CERTAIN.
> I am never certain before coffee.
HOW LONG HAVE YOU BEEN NEVER CERTAIN
BEFORE COFFEE?
> Since I started writing firmware.
TELL ME MORE ABOUT WRITING FIRMWARE.
> _
```

fs0307 - ELIZA-style chat

```
       .-----.
       | o o |
       |  -  |
       '--+--'
      /|     |\
       |_____|

     TINY ENGINEER
   booting desk robot...
   [##########--------]
```

fs0307 - Tiny Engineer splash

```
draw or glue your screen inside
the dashed line (visible area)
```

fs0307 - Blank template

```
~/tiny-engineer$ pio run
Processing esp32-c3-devkitm-1 (espressif32)
-------------------------------------------
Compiling .pio/build/src/anim/poses.cpp.o
Compiling .pio/build/src/http/routes.cpp.o
Compiling .pio/build/src/servos/park.cpp.o
Linking .pio/build/firmware.elf
Checking size .pio/build/firmware.elf
RAM:   [==        ]  17.4% (57012 bytes)
Flash: [=====     ]  52.1% (1023890 bytes)
Building .pio/build/firmware.bin
========== [SUCCESS] Took 12.84 seconds ==========
~/tiny-engineer$ _
```

fs0307 - Build output

```
// wave hello, then get back to work
void greet(Robot &bot) {
  for (int i = 0; i < 3; i++) {
    bot.arm(Side::Right, 0.8f);
    delay(180);
    bot.arm(Side::Right, -0.2f);
    delay(180);
  }
  bot.say("Hello, world!");
  bot.pose("typing");
}
```

fs0307 - Source code

```
> fix the failing servo test

* Reading test/servos/test_park.cpp
* The test expects the elbow to park at
  0.0, but the default is now -0.1.
* Updating the expected value...
  - TEST_ASSERT_EQUAL_FLOAT(0.0f, pose);
  + TEST_ASSERT_EQUAL_FLOAT(-0.1f, pose);
* Ran pio test -e native: 42 passed

> _
```

fs0307 - AI agent chat

Tiny Engineer - laptop screen inserts: Colour screens

Servo preset: EMAX ES9051 (es9051)

Print only one page for es9051 parts.

Colour or black-only printer on white paper. Dark fills hide the paper colour.

Insert 37.2 x 28.4 mm (cut on the grey line). Visible window 36.8 x 19.9 mm, 0.2 mm from the left and 1.7 mm from the top edge.

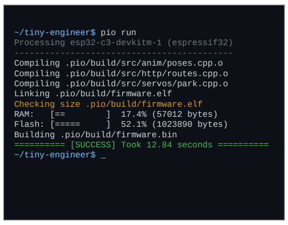
Slide the paper insert, then a same-size piece of clear PET, into the lid pocket in front of LaptopScreen.

10 mm

0 10 20 30 40 50

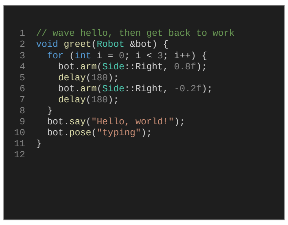
Square must measure 10 x 10 mm and the ruler 50 mm.

If not, reprint at 100% / Actual size (scaling off).



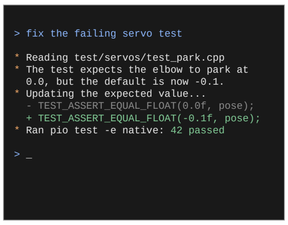
```
~/tiny-engineer$ pio run
Processing esp32-c3-devkitm-1 (espressif32)
----------------------------------------
Compiling .pio/build/src/anim/poses.cpp.o
Compiling .pio/build/src/http/routes.cpp.o
Compiling .pio/build/src/servos/park.cpp.o
Linking .pio/build/firmware.elf
Checking size .pio/build/firmware.elf
RAM:   [==        ]  17.4% (57012 bytes)
Flash: [=====     ]  52.1% (1023890 bytes)
Building .pio/build/firmware.bin
========== [SUCCESS] Took 12.84 seconds ==========
~/tiny-engineer$ _
```

es9051 - Build output



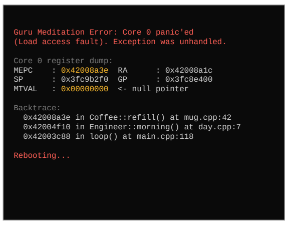
```
// wave hello, then get back to work
void greet(Robot &bot) {
  for (int i = 0; i < 3; i++) {
    bot.arm(Side::Right, 0.8f);
    delay(180);
    bot.arm(Side::Right, -0.2f);
    delay(180);
  }
  bot.say("Hello, world!");
  bot.pose("typing");
}
```

es9051 - Source code



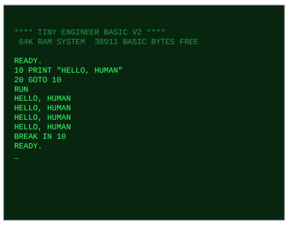
```
> fix the failing servo test

* Reading test/servos/test_park.cpp
* The test expects the elbow to park at
  0.0, but the default is now -0.1.
* Updating the expected value...
  - TEST_ASSERT_EQUAL_FLOAT(0.0f, pose);
  + TEST_ASSERT_EQUAL_FLOAT(-0.1f, pose);
* Ran pio test -e native: 42 passed

> _
```

es9051 - AI agent chat



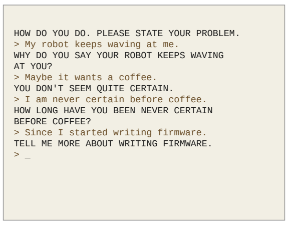
```
Guru Meditation Error: Core 0 panic'ed
(Load access fault). Exception was unhandled.

Core 0 register dump:
MEPC    : 0x42008a3e  RA      : 0x42008a1c
SP      : 0x3fc9b2f0  GP      : 0x3fc8e400
MTVAL   : 0x00000000  <- null pointer

Backtrace:
  0x42008a3e in Coffee::refill() at mug.cpp:42
  0x42004f10 in Engineer::morning() at day.cpp:7
  0x42003c88 in loop() at main.cpp:118

Rebooting...
```

es9051 - Crash / stack trace



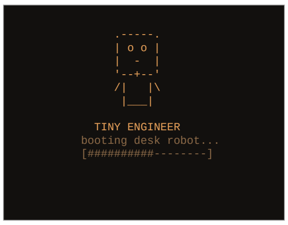
```
**** TINY ENGINEER BASIC V2 ****
 64K RAM SYSTEM  38911 BASIC BYTES FREE

READY.
10 PRINT "HELLO, HUMAN"
20 GOTO 10
RUN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
BREAK IN 10
READY.
_
```

es9051 - Retro computer



```
HOW DO YOU DO. PLEASE STATE YOUR PROBLEM.
> My robot keeps waving at me.
WHY DO YOU SAY YOUR ROBOT KEEPS WAVING
AT YOU?
> Maybe it wants a coffee.
YOU DON'T SEEM QUITE CERTAIN.
> I am never certain before coffee.
HOW LONG HAVE YOU BEEN NEVER CERTAIN
BEFORE COFFEE?
> Since I started writing firmware.
TELL ME MORE ABOUT WRITING FIRMWARE.
> _
```

es9051 - ELIZA-style chat

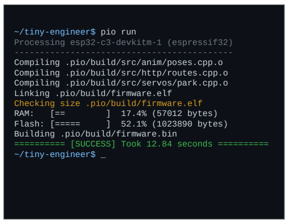


```
      .-----.
      | o o |
      |  -  |
      '--+--'
      /|   |\
       |___|

    TINY ENGINEER
  booting desk robot...
  [##########--------]
```

es9051 - Tiny Engineer splash

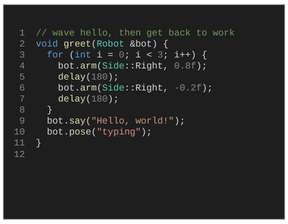
draw or glue your screen inside the dashed line (visible area)

es9051 - Blank template



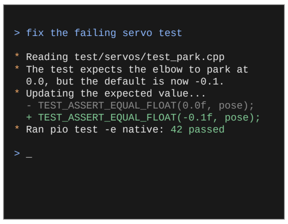
```
~/tiny-engineer$ pio run
Processing esp32-c3-devkitm-1 (espressif32)
----------------------------------------
Compiling .pio/build/src/anim/poses.cpp.o
Compiling .pio/build/src/http/routes.cpp.o
Compiling .pio/build/src/servos/park.cpp.o
Linking .pio/build/firmware.elf
Checking size .pio/build/firmware.elf
RAM:   [==        ]  17.4% (57012 bytes)
Flash: [=====     ]  52.1% (1023890 bytes)
Building .pio/build/firmware.bin
========== [SUCCESS] Took 12.84 seconds ==========
~/tiny-engineer$ _
```

es9051 - Build output



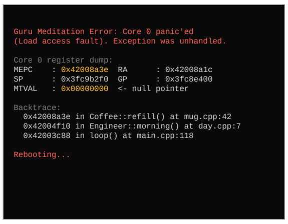
```
// wave hello, then get back to work
void greet(Robot &bot) {
  for (int i = 0; i < 3; i++) {
    bot.arm(Side::Right, 0.8f);
    delay(180);
    bot.arm(Side::Right, -0.2f);
    delay(180);
  }
  bot.say("Hello, world!");
  bot.pose("typing");
}
```

es9051 - Source code



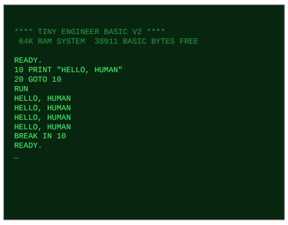
```
> fix the failing servo test

* Reading test/servos/test_park.cpp
* The test expects the elbow to park at
  0.0, but the default is now -0.1.
* Updating the expected value...
  - TEST_ASSERT_EQUAL_FLOAT(0.0f, pose);
  + TEST_ASSERT_EQUAL_FLOAT(-0.1f, pose);
* Ran pio test -e native: 42 passed

> _
```

es9051 - AI agent chat



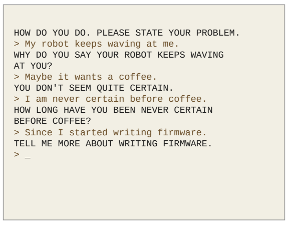
```
Guru Meditation Error: Core 0 panic'ed
(Load access fault). Exception was unhandled.

Core 0 register dump:
MEPC    : 0x42008a3e  RA      : 0x42008a1c
SP      : 0x3fc9b2f0  GP      : 0x3fc8e400
MTVAL   : 0x00000000  <- null pointer

Backtrace:
  0x42008a3e in Coffee::refill() at mug.cpp:42
  0x42004f10 in Engineer::morning() at day.cpp:7
  0x42003c88 in loop() at main.cpp:118

Rebooting...
```

es9051 - Crash / stack trace



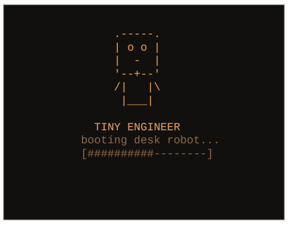
```
**** TINY ENGINEER BASIC V2 ****
 64K RAM SYSTEM  38911 BASIC BYTES FREE

READY.
10 PRINT "HELLO, HUMAN"
20 GOTO 10
RUN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
BREAK IN 10
READY.
_
```

es9051 - Retro computer



```
HOW DO YOU DO. PLEASE STATE YOUR PROBLEM.
> My robot keeps waving at me.
WHY DO YOU SAY YOUR ROBOT KEEPS WAVING
AT YOU?
> Maybe it wants a coffee.
YOU DON'T SEEM QUITE CERTAIN.
> I am never certain before coffee.
HOW LONG HAVE YOU BEEN NEVER CERTAIN
BEFORE COFFEE?
> Since I started writing firmware.
TELL ME MORE ABOUT WRITING FIRMWARE.
> _
```

es9051 - ELIZA-style chat

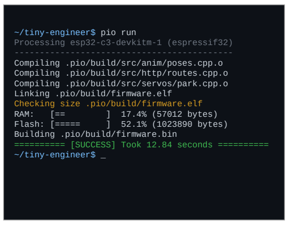


```
      .-----.
      | o o |
      |  -  |
      '--+--'
      /|   |\
       |___|

    TINY ENGINEER
  booting desk robot...
  [##########--------]
```

es9051 - Tiny Engineer splash

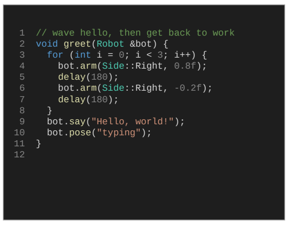
draw or glue your screen inside the dashed line (visible area)

es9051 - Blank template



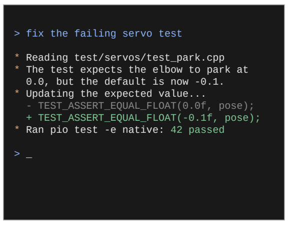
```
~/tiny-engineer$ pio run
Processing esp32-c3-devkitm-1 (espressif32)
----------------------------------------
Compiling .pio/build/src/anim/poses.cpp.o
Compiling .pio/build/src/http/routes.cpp.o
Compiling .pio/build/src/servos/park.cpp.o
Linking .pio/build/firmware.elf
Checking size .pio/build/firmware.elf
RAM:   [==        ]  17.4% (57012 bytes)
Flash: [=====     ]  52.1% (1023890 bytes)
Building .pio/build/firmware.bin
========== [SUCCESS] Took 12.84 seconds ==========
~/tiny-engineer$ _
```

es9051 - Build output



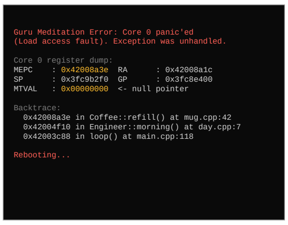
```
// wave hello, then get back to work
void greet(Robot &bot) {
  for (int i = 0; i < 3; i++) {
    bot.arm(Side::Right, 0.8f);
    delay(180);
    bot.arm(Side::Right, -0.2f);
    delay(180);
  }
  bot.say("Hello, world!");
  bot.pose("typing");
}
```

es9051 - Source code



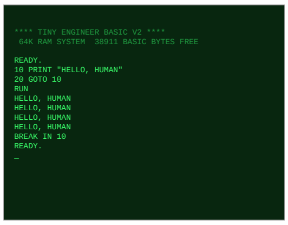
```
> fix the failing servo test

* Reading test/servos/test_park.cpp
* The test expects the elbow to park at
  0.0, but the default is now -0.1.
* Updating the expected value...
  - TEST_ASSERT_EQUAL_FLOAT(0.0f, pose);
  + TEST_ASSERT_EQUAL_FLOAT(-0.1f, pose);
* Ran pio test -e native: 42 passed

> _
```

es9051 - AI agent chat



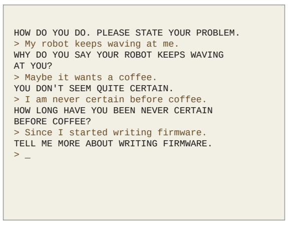
```
Guru Meditation Error: Core 0 panic'ed
(Load access fault). Exception was unhandled.

Core 0 register dump:
MEPC    : 0x42008a3e  RA      : 0x42008a1c
SP      : 0x3fc9b2f0  GP      : 0x3fc8e400
MTVAL   : 0x00000000  <- null pointer

Backtrace:
  0x42008a3e in Coffee::refill() at mug.cpp:42
  0x42004f10 in Engineer::morning() at day.cpp:7
  0x42003c88 in loop() at main.cpp:118

Rebooting...
```

es9051 - Crash / stack trace



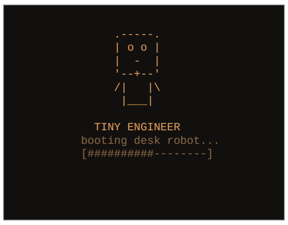
```
**** TINY ENGINEER BASIC V2 ****
 64K RAM SYSTEM  38911 BASIC BYTES FREE

READY.
10 PRINT "HELLO, HUMAN"
20 GOTO 10
RUN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
BREAK IN 10
READY.
_
```

es9051 - Retro computer



```
HOW DO YOU DO. PLEASE STATE YOUR PROBLEM.
> My robot keeps waving at me.
WHY DO YOU SAY YOUR ROBOT KEEPS WAVING
AT YOU?
> Maybe it wants a coffee.
YOU DON'T SEEM QUITE CERTAIN.
> I am never certain before coffee.
HOW LONG HAVE YOU BEEN NEVER CERTAIN
BEFORE COFFEE?
> Since I started writing firmware.
TELL ME MORE ABOUT WRITING FIRMWARE.
> _
```

es9051 - ELIZA-style chat



```
      .-----.
      | o o |
      |  -  |
      '--+--'
      /|   |\
       |___|

    TINY ENGINEER
  booting desk robot...
  [##########--------]
```

es9051 - Tiny Engineer splash

draw or glue your screen inside the dashed line (visible area)

es9051 - Blank template

Tiny Engineer - laptop screen inserts: Paper screens

Servo preset: EMAX ES9051 (es9051)

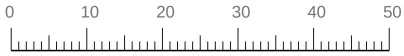
Print only one page for es9051 parts.

Any printer, any paper colour: no fill, so the paper becomes the screen.

Insert 37.2 x 28.4 mm (cut on the grey line). Visible window 36.8 x 19.9 mm, 0.2 mm from the left and 1.7 mm from the top edge.

Slide the paper insert, then a same-size piece of clear PET, into the lid pocket in front of LaptopScreen.

10 mm



Square must measure 10 x 10 mm and the ruler 50 mm.

If not, reprint at 100% / Actual size (scaling off).

```
~/tiny-engineer$ pio run
Processing esp32-c3-devkitm-1 (espressif32)
------------------------------------------
Compiling .pio/build/src/anim/poses.cpp.o
Compiling .pio/build/src/http/routes.cpp.o
Compiling .pio/build/src/servos/park.cpp.o
Linking .pio/build/firmware.elf
Checking size .pio/build/firmware.elf
RAM:   [==        ]  17.4% (57012 bytes)
Flash: [=====     ]  52.1% (1023890 bytes)
Building .pio/build/firmware.bin
========== [SUCCESS] Took 12.84 seconds ==========
~/tiny-engineer$ _
```

es9051 - Build output

```
// wave hello, then get back to work
void greet(Robot &bot) {
  for (int i = 0; i < 3; i++) {
    bot.arm(Side::Right, 0.8f);
    delay(180);
    bot.arm(Side::Right, -0.2f);
    delay(180);
  }
  bot.say("Hello, world!");
  bot.pose("typing");
}
```

es9051 - Source code

```
> fix the failing servo test

* Reading test/servos/test_park.cpp
* The test expects the elbow to park at
  0.0, but the default is now -0.1.
* Updating the expected value...
  - TEST_ASSERT_EQUAL_FLOAT(0.0f, pose);
  + TEST_ASSERT_EQUAL_FLOAT(-0.1f, pose);
* Ran pio test -e native: 42 passed

> _
```

es9051 - AI agent chat

```
Guru Meditation Error: Core 0 panic'ed
(Load access fault). Exception was unhandled.

Core 0 register dump:
MEPC    : 0x42008a3e  RA      : 0x42008a1c
SP      : 0x3fc9b2f0  GP      : 0x3fc8e400
MTVAL   : 0x00000000  <- null pointer

Backtrace:
  0x42008a3e in Coffee::refill() at mug.cpp:42
  0x42004f10 in Engineer::morning() at day.cpp:7
  0x42003c88 in loop() at main.cpp:118

Rebooting...
```

es9051 - Crash / stack trace

```
**** TINY ENGINEER BASIC V2 ****
 64K RAM SYSTEM  38911 BASIC BYTES FREE

READY.
10 PRINT "HELLO, HUMAN"
20 GOTO 10
RUN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
BREAK IN 10
READY.
_
```

es9051 - Retro computer

```
HOW DO YOU DO. PLEASE STATE YOUR PROBLEM.
> My robot keeps waving at me.
WHY DO YOU SAY YOUR ROBOT KEEPS WAVING
AT YOU?
> Maybe it wants a coffee.
YOU DON'T SEEM QUITE CERTAIN.
> I am never certain before coffee.
HOW LONG HAVE YOU BEEN NEVER CERTAIN
BEFORE COFFEE?
> Since I started writing firmware.
TELL ME MORE ABOUT WRITING FIRMWARE.
> _
```

es9051 - ELIZA-style chat

```
     .-----.
     | o o |
     |  -  |
     '--+--'
     /|   |\
      |___|

  TINY ENGINEER
booting desk robot...
[##########--------]
```

es9051 - Tiny Engineer splash

draw or glue your screen inside
the dashed line (visible area)

es9051 - Blank template

```
~/tiny-engineer$ pio run
Processing esp32-c3-devkitm-1 (espressif32)
------------------------------------------
Compiling .pio/build/src/anim/poses.cpp.o
Compiling .pio/build/src/http/routes.cpp.o
Compiling .pio/build/src/servos/park.cpp.o
Linking .pio/build/firmware.elf
Checking size .pio/build/firmware.elf
RAM:   [==        ]  17.4% (57012 bytes)
Flash: [=====     ]  52.1% (1023890 bytes)
Building .pio/build/firmware.bin
========== [SUCCESS] Took 12.84 seconds ==========
~/tiny-engineer$ _
```

es9051 - Build output

```
// wave hello, then get back to work
void greet(Robot &bot) {
  for (int i = 0; i < 3; i++) {
    bot.arm(Side::Right, 0.8f);
    delay(180);
    bot.arm(Side::Right, -0.2f);
    delay(180);
  }
  bot.say("Hello, world!");
  bot.pose("typing");
}
```

es9051 - Source code

```
> fix the failing servo test

* Reading test/servos/test_park.cpp
* The test expects the elbow to park at
  0.0, but the default is now -0.1.
* Updating the expected value...
  - TEST_ASSERT_EQUAL_FLOAT(0.0f, pose);
  + TEST_ASSERT_EQUAL_FLOAT(-0.1f, pose);
* Ran pio test -e native: 42 passed

> _
```

es9051 - AI agent chat

```
Guru Meditation Error: Core 0 panic'ed
(Load access fault). Exception was unhandled.

Core 0 register dump:
MEPC    : 0x42008a3e  RA      : 0x42008a1c
SP      : 0x3fc9b2f0  GP      : 0x3fc8e400
MTVAL   : 0x00000000  <- null pointer

Backtrace:
  0x42008a3e in Coffee::refill() at mug.cpp:42
  0x42004f10 in Engineer::morning() at day.cpp:7
  0x42003c88 in loop() at main.cpp:118

Rebooting...
```

es9051 - Crash / stack trace

```
**** TINY ENGINEER BASIC V2 ****
 64K RAM SYSTEM  38911 BASIC BYTES FREE

READY.
10 PRINT "HELLO, HUMAN"
20 GOTO 10
RUN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
BREAK IN 10
READY.
_
```

es9051 - Retro computer

```
HOW DO YOU DO. PLEASE STATE YOUR PROBLEM.
> My robot keeps waving at me.
WHY DO YOU SAY YOUR ROBOT KEEPS WAVING
AT YOU?
> Maybe it wants a coffee.
YOU DON'T SEEM QUITE CERTAIN.
> I am never certain before coffee.
HOW LONG HAVE YOU BEEN NEVER CERTAIN
BEFORE COFFEE?
> Since I started writing firmware.
TELL ME MORE ABOUT WRITING FIRMWARE.
> _
```

es9051 - ELIZA-style chat

```
     .-----.
     | o o |
     |  -  |
     '--+--'
     /|   |\
      |___|

  TINY ENGINEER
booting desk robot...
[##########--------]
```

es9051 - Tiny Engineer splash

draw or glue your screen inside
the dashed line (visible area)

es9051 - Blank template

```
~/tiny-engineer$ pio run
Processing esp32-c3-devkitm-1 (espressif32)
------------------------------------------
Compiling .pio/build/src/anim/poses.cpp.o
Compiling .pio/build/src/http/routes.cpp.o
Compiling .pio/build/src/servos/park.cpp.o
Linking .pio/build/firmware.elf
Checking size .pio/build/firmware.elf
RAM:   [==        ]  17.4% (57012 bytes)
Flash: [=====     ]  52.1% (1023890 bytes)
Building .pio/build/firmware.bin
========== [SUCCESS] Took 12.84 seconds ==========
~/tiny-engineer$ _
```

es9051 - Build output

```
// wave hello, then get back to work
void greet(Robot &bot) {
  for (int i = 0; i < 3; i++) {
    bot.arm(Side::Right, 0.8f);
    delay(180);
    bot.arm(Side::Right, -0.2f);
    delay(180);
  }
  bot.say("Hello, world!");
  bot.pose("typing");
}
```

es9051 - Source code

```
> fix the failing servo test

* Reading test/servos/test_park.cpp
* The test expects the elbow to park at
  0.0, but the default is now -0.1.
* Updating the expected value...
  - TEST_ASSERT_EQUAL_FLOAT(0.0f, pose);
  + TEST_ASSERT_EQUAL_FLOAT(-0.1f, pose);
* Ran pio test -e native: 42 passed

> _
```

es9051 - AI agent chat

```
Guru Meditation Error: Core 0 panic'ed
(Load access fault). Exception was unhandled.

Core 0 register dump:
MEPC    : 0x42008a3e  RA      : 0x42008a1c
SP      : 0x3fc9b2f0  GP      : 0x3fc8e400
MTVAL   : 0x00000000  <- null pointer

Backtrace:
  0x42008a3e in Coffee::refill() at mug.cpp:42
  0x42004f10 in Engineer::morning() at day.cpp:7
  0x42003c88 in loop() at main.cpp:118

Rebooting...
```

es9051 - Crash / stack trace

```
**** TINY ENGINEER BASIC V2 ****
 64K RAM SYSTEM  38911 BASIC BYTES FREE

READY.
10 PRINT "HELLO, HUMAN"
20 GOTO 10
RUN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
BREAK IN 10
READY.
_
```

es9051 - Retro computer

```
HOW DO YOU DO. PLEASE STATE YOUR PROBLEM.
> My robot keeps waving at me.
WHY DO YOU SAY YOUR ROBOT KEEPS WAVING
AT YOU?
> Maybe it wants a coffee.
YOU DON'T SEEM QUITE CERTAIN.
> I am never certain before coffee.
HOW LONG HAVE YOU BEEN NEVER CERTAIN
BEFORE COFFEE?
> Since I started writing firmware.
TELL ME MORE ABOUT WRITING FIRMWARE.
> _
```

es9051 - ELIZA-style chat

```
     .-----.
     | o o |
     |  -  |
     '--+--'
     /|   |\
      |___|

  TINY ENGINEER
booting desk robot...
[##########--------]
```

es9051 - Tiny Engineer splash

draw or glue your screen inside
the dashed line (visible area)

es9051 - Blank template

Tiny Engineer - laptop screen inserts: Colour screens

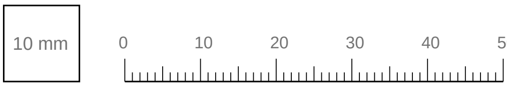
Servo preset: PowerHD HD-1370A (hd1370a)

Print only one page for hd1370a parts.

Colour or black-only printer on white paper. Dark fills hide the paper colour.

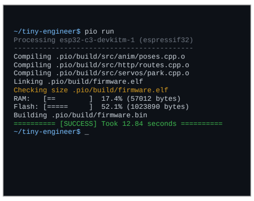
Insert 32.7 x 25.2 mm (cut on the grey line). Visible window 32.3 x 16.7 mm, 0.2 mm from the left and 1.7 mm from the top edge.

Slide the paper insert, then a same-size piece of clear PET, into the lid pocket in front of LaptopScreen.



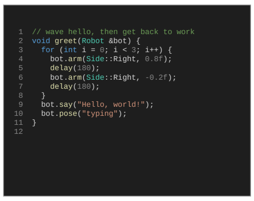
Square must measure 10 x 10 mm and the ruler 50 mm.

If not, reprint at 100% / Actual size (scaling off).



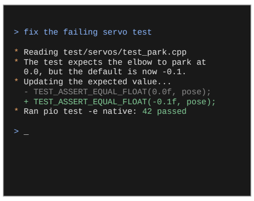
```
~/tiny-engineer$ pio run
Processing esp32-c3-devkitm-1 (espressif32)
-------------------------------------------
Compiling .pio/build/src/anim/poses.cpp.o
Compiling .pio/build/src/http/routes.cpp.o
Compiling .pio/build/src/servos/park.cpp.o
Linking .pio/build/firmware.elf
Checking size .pio/build/firmware.elf
RAM:   [==        ]  17.4% (57012 bytes)
Flash: [=====     ]  52.1% (1023890 bytes)
Building .pio/build/firmware.bin
========== [SUCCESS] Took 12.84 seconds ==========
~/tiny-engineer$ _
```

hd1370a - Build output



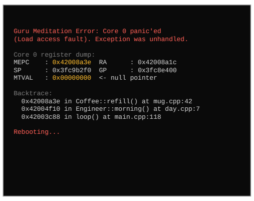
```
// wave hello, then get back to work
void greet(Robot &bot) {
  for (int i = 0; i < 3; i++) {
    bot.arm(Side::Right, 0.8f);
    delay(180);
    bot.arm(Side::Right, -0.2f);
    delay(180);
  }
  bot.say("Hello, world!");
  bot.pose("typing");
}
```

hd1370a - Source code



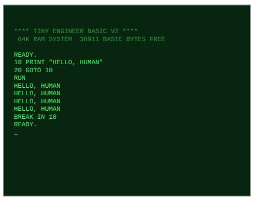
```
> fix the failing servo test

* Reading test/servos/test_park.cpp
* The test expects the elbow to park at
  0.0, but the default is now -0.1.
* Updating the expected value...
  - TEST_ASSERT_EQUAL_FLOAT(0.0f, pose);
  + TEST_ASSERT_EQUAL_FLOAT(-0.1f, pose);
* Ran pio test -e native: 42 passed

> _
```

hd1370a - AI agent chat



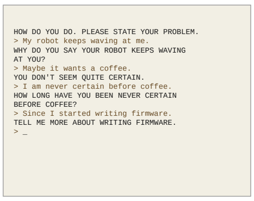
```
Guru Meditation Error: Core 0 panic'ed
(Load access fault). Exception was unhandled.

Core 0 register dump:
MEPC    : 0x42008a3e  RA      : 0x42008a1c
SP      : 0x3fc9b2f0  GP      : 0x3fc8e400
MTVAL   : 0x00000000  <- null pointer

Backtrace:
  0x42008a3e in Coffee::refill() at mug.cpp:42
  0x42004f10 in Engineer::morning() at day.cpp:7
  0x42003c88 in loop() at main.cpp:118

Rebooting...
```

hd1370a - Crash / stack trace



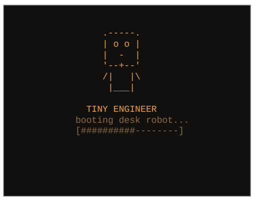
```
**** TINY ENGINEER BASIC V2 ****
 64K RAM SYSTEM  38911 BASIC BYTES FREE

READY.
10 PRINT "HELLO, HUMAN"
20 GOTO 10
RUN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
BREAK IN 10
READY.
_
```

hd1370a - Retro computer



```
HOW DO YOU DO. PLEASE STATE YOUR PROBLEM.
> My robot keeps waving at me.
WHY DO YOU SAY YOUR ROBOT KEEPS WAVING
AT YOU?
> Maybe it wants a coffee.
YOU DON'T SEEM QUITE CERTAIN.
> I am never certain before coffee.
HOW LONG HAVE YOU BEEN NEVER CERTAIN
BEFORE COFFEE?
> Since I started writing firmware.
TELL ME MORE ABOUT WRITING FIRMWARE.
> _
```

hd1370a - ELIZA-style chat

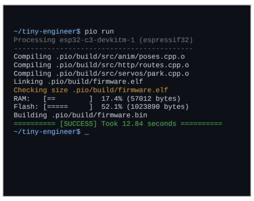


```
      .-----.
      | o o |
      |  -  |
      '--+--'
     /|     |\
      |_____|

    TINY ENGINEER
  booting desk robot...
  [##########--------]
```

hd1370a - Tiny Engineer splash

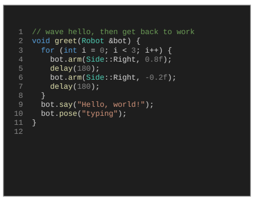
draw or glue your screen inside
the dashed line (visible area)

hd1370a - Blank template



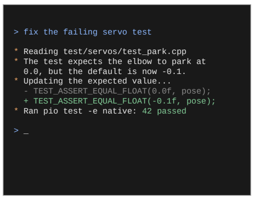
```
~/tiny-engineer$ pio run
Processing esp32-c3-devkitm-1 (espressif32)
-------------------------------------------
Compiling .pio/build/src/anim/poses.cpp.o
Compiling .pio/build/src/http/routes.cpp.o
Compiling .pio/build/src/servos/park.cpp.o
Linking .pio/build/firmware.elf
Checking size .pio/build/firmware.elf
RAM:   [==        ]  17.4% (57012 bytes)
Flash: [=====     ]  52.1% (1023890 bytes)
Building .pio/build/firmware.bin
========== [SUCCESS] Took 12.84 seconds ==========
~/tiny-engineer$ _
```

hd1370a - Build output



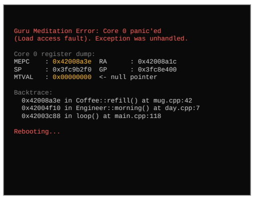
```
// wave hello, then get back to work
void greet(Robot &bot) {
  for (int i = 0; i < 3; i++) {
    bot.arm(Side::Right, 0.8f);
    delay(180);
    bot.arm(Side::Right, -0.2f);
    delay(180);
  }
  bot.say("Hello, world!");
  bot.pose("typing");
}
```

hd1370a - Source code



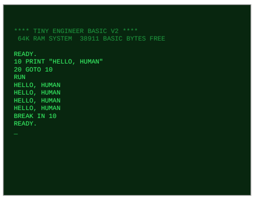
```
> fix the failing servo test

* Reading test/servos/test_park.cpp
* The test expects the elbow to park at
  0.0, but the default is now -0.1.
* Updating the expected value...
  - TEST_ASSERT_EQUAL_FLOAT(0.0f, pose);
  + TEST_ASSERT_EQUAL_FLOAT(-0.1f, pose);
* Ran pio test -e native: 42 passed

> _
```

hd1370a - AI agent chat



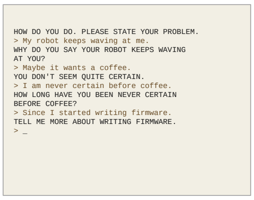
```
Guru Meditation Error: Core 0 panic'ed
(Load access fault). Exception was unhandled.

Core 0 register dump:
MEPC    : 0x42008a3e  RA      : 0x42008a1c
SP      : 0x3fc9b2f0  GP      : 0x3fc8e400
MTVAL   : 0x00000000  <- null pointer

Backtrace:
  0x42008a3e in Coffee::refill() at mug.cpp:42
  0x42004f10 in Engineer::morning() at day.cpp:7
  0x42003c88 in loop() at main.cpp:118

Rebooting...
```

hd1370a - Crash / stack trace



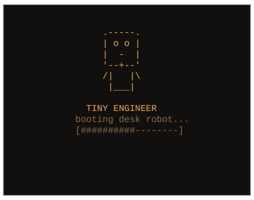
```
**** TINY ENGINEER BASIC V2 ****
 64K RAM SYSTEM  38911 BASIC BYTES FREE

READY.
10 PRINT "HELLO, HUMAN"
20 GOTO 10
RUN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
BREAK IN 10
READY.
_
```

hd1370a - Retro computer



```
HOW DO YOU DO. PLEASE STATE YOUR PROBLEM.
> My robot keeps waving at me.
WHY DO YOU SAY YOUR ROBOT KEEPS WAVING
AT YOU?
> Maybe it wants a coffee.
YOU DON'T SEEM QUITE CERTAIN.
> I am never certain before coffee.
HOW LONG HAVE YOU BEEN NEVER CERTAIN
BEFORE COFFEE?
> Since I started writing firmware.
TELL ME MORE ABOUT WRITING FIRMWARE.
> _
```

hd1370a - ELIZA-style chat

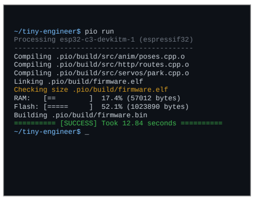


```
      .-----.
      | o o |
      |  -  |
      '--+--'
     /|     |\
      |_____|

    TINY ENGINEER
  booting desk robot...
  [##########--------]
```

hd1370a - Tiny Engineer splash

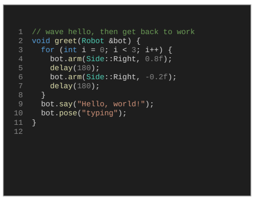
draw or glue your screen inside
the dashed line (visible area)

hd1370a - Blank template



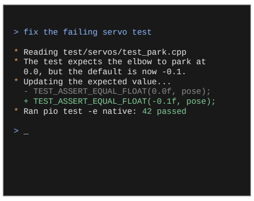
```
~/tiny-engineer$ pio run
Processing esp32-c3-devkitm-1 (espressif32)
-------------------------------------------
Compiling .pio/build/src/anim/poses.cpp.o
Compiling .pio/build/src/http/routes.cpp.o
Compiling .pio/build/src/servos/park.cpp.o
Linking .pio/build/firmware.elf
Checking size .pio/build/firmware.elf
RAM:   [==        ]  17.4% (57012 bytes)
Flash: [=====     ]  52.1% (1023890 bytes)
Building .pio/build/firmware.bin
========== [SUCCESS] Took 12.84 seconds ==========
~/tiny-engineer$ _
```

hd1370a - Build output



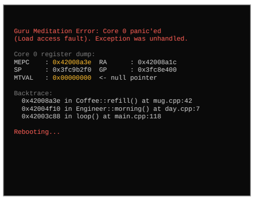
```
// wave hello, then get back to work
void greet(Robot &bot) {
  for (int i = 0; i < 3; i++) {
    bot.arm(Side::Right, 0.8f);
    delay(180);
    bot.arm(Side::Right, -0.2f);
    delay(180);
  }
  bot.say("Hello, world!");
  bot.pose("typing");
}
```

hd1370a - Source code



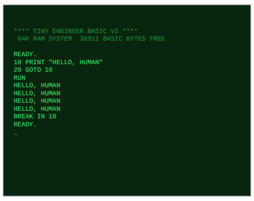
```
> fix the failing servo test

* Reading test/servos/test_park.cpp
* The test expects the elbow to park at
  0.0, but the default is now -0.1.
* Updating the expected value...
  - TEST_ASSERT_EQUAL_FLOAT(0.0f, pose);
  + TEST_ASSERT_EQUAL_FLOAT(-0.1f, pose);
* Ran pio test -e native: 42 passed

> _
```

hd1370a - AI agent chat



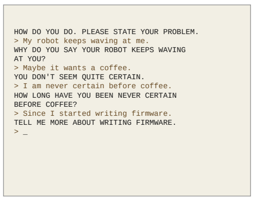
```
Guru Meditation Error: Core 0 panic'ed
(Load access fault). Exception was unhandled.

Core 0 register dump:
MEPC    : 0x42008a3e  RA      : 0x42008a1c
SP      : 0x3fc9b2f0  GP      : 0x3fc8e400
MTVAL   : 0x00000000  <- null pointer

Backtrace:
  0x42008a3e in Coffee::refill() at mug.cpp:42
  0x42004f10 in Engineer::morning() at day.cpp:7
  0x42003c88 in loop() at main.cpp:118

Rebooting...
```

hd1370a - Crash / stack trace



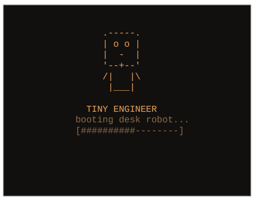
```
**** TINY ENGINEER BASIC V2 ****
 64K RAM SYSTEM  38911 BASIC BYTES FREE

READY.
10 PRINT "HELLO, HUMAN"
20 GOTO 10
RUN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
BREAK IN 10
READY.
_
```

hd1370a - Retro computer



```
HOW DO YOU DO. PLEASE STATE YOUR PROBLEM.
> My robot keeps waving at me.
WHY DO YOU SAY YOUR ROBOT KEEPS WAVING
AT YOU?
> Maybe it wants a coffee.
YOU DON'T SEEM QUITE CERTAIN.
> I am never certain before coffee.
HOW LONG HAVE YOU BEEN NEVER CERTAIN
BEFORE COFFEE?
> Since I started writing firmware.
TELL ME MORE ABOUT WRITING FIRMWARE.
> _
```

hd1370a - ELIZA-style chat

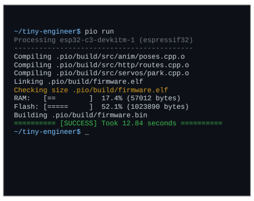


```
      .-----.
      | o o |
      |  -  |
      '--+--'
     /|     |\
      |_____|

    TINY ENGINEER
  booting desk robot...
  [##########--------]
```

hd1370a - Tiny Engineer splash

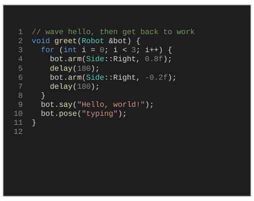
draw or glue your screen inside
the dashed line (visible area)

hd1370a - Blank template



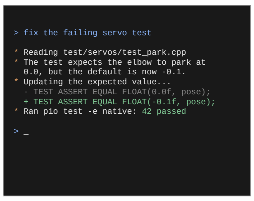
```
~/tiny-engineer$ pio run
Processing esp32-c3-devkitm-1 (espressif32)
-------------------------------------------
Compiling .pio/build/src/anim/poses.cpp.o
Compiling .pio/build/src/http/routes.cpp.o
Compiling .pio/build/src/servos/park.cpp.o
Linking .pio/build/firmware.elf
Checking size .pio/build/firmware.elf
RAM:   [==        ]  17.4% (57012 bytes)
Flash: [=====     ]  52.1% (1023890 bytes)
Building .pio/build/firmware.bin
========== [SUCCESS] Took 12.84 seconds ==========
~/tiny-engineer$ _
```

hd1370a - Build output



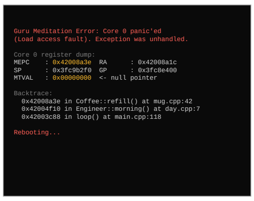
```
// wave hello, then get back to work
void greet(Robot &bot) {
  for (int i = 0; i < 3; i++) {
    bot.arm(Side::Right, 0.8f);
    delay(180);
    bot.arm(Side::Right, -0.2f);
    delay(180);
  }
  bot.say("Hello, world!");
  bot.pose("typing");
}
```

hd1370a - Source code



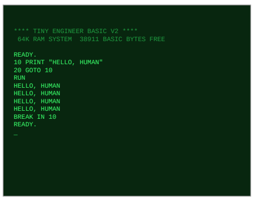
```
> fix the failing servo test

* Reading test/servos/test_park.cpp
* The test expects the elbow to park at
  0.0, but the default is now -0.1.
* Updating the expected value...
  - TEST_ASSERT_EQUAL_FLOAT(0.0f, pose);
  + TEST_ASSERT_EQUAL_FLOAT(-0.1f, pose);
* Ran pio test -e native: 42 passed

> _
```

hd1370a - AI agent chat



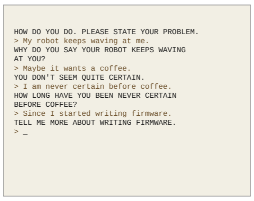
```
Guru Meditation Error: Core 0 panic'ed
(Load access fault). Exception was unhandled.

Core 0 register dump:
MEPC    : 0x42008a3e  RA      : 0x42008a1c
SP      : 0x3fc9b2f0  GP      : 0x3fc8e400
MTVAL   : 0x00000000  <- null pointer

Backtrace:
  0x42008a3e in Coffee::refill() at mug.cpp:42
  0x42004f10 in Engineer::morning() at day.cpp:7
  0x42003c88 in loop() at main.cpp:118

Rebooting...
```

hd1370a - Crash / stack trace



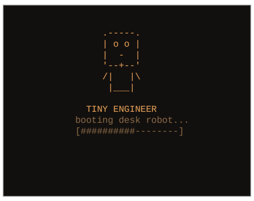
```
**** TINY ENGINEER BASIC V2 ****
 64K RAM SYSTEM  38911 BASIC BYTES FREE

READY.
10 PRINT "HELLO, HUMAN"
20 GOTO 10
RUN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
BREAK IN 10
READY.
_
```

hd1370a - Retro computer



```
HOW DO YOU DO. PLEASE STATE YOUR PROBLEM.
> My robot keeps waving at me.
WHY DO YOU SAY YOUR ROBOT KEEPS WAVING
AT YOU?
> Maybe it wants a coffee.
YOU DON'T SEEM QUITE CERTAIN.
> I am never certain before coffee.
HOW LONG HAVE YOU BEEN NEVER CERTAIN
BEFORE COFFEE?
> Since I started writing firmware.
TELL ME MORE ABOUT WRITING FIRMWARE.
> _
```

hd1370a - ELIZA-style chat

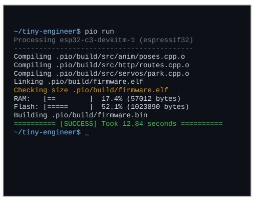


```
      .-----.
      | o o |
      |  -  |
      '--+--'
     /|     |\
      |_____|

    TINY ENGINEER
  booting desk robot...
  [##########--------]
```

hd1370a - Tiny Engineer splash

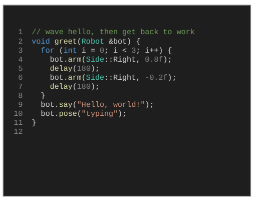
draw or glue your screen inside
the dashed line (visible area)

hd1370a - Blank template



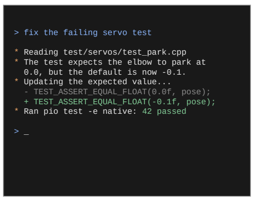
```
~/tiny-engineer$ pio run
Processing esp32-c3-devkitm-1 (espressif32)
-------------------------------------------
Compiling .pio/build/src/anim/poses.cpp.o
Compiling .pio/build/src/http/routes.cpp.o
Compiling .pio/build/src/servos/park.cpp.o
Linking .pio/build/firmware.elf
Checking size .pio/build/firmware.elf
RAM:   [==        ]  17.4% (57012 bytes)
Flash: [=====     ]  52.1% (1023890 bytes)
Building .pio/build/firmware.bin
========== [SUCCESS] Took 12.84 seconds ==========
~/tiny-engineer$ _
```

hd1370a - Build output



```
// wave hello, then get back to work
void greet(Robot &bot) {
  for (int i = 0; i < 3; i++) {
    bot.arm(Side::Right, 0.8f);
    delay(180);
    bot.arm(Side::Right, -0.2f);
    delay(180);
  }
  bot.say("Hello, world!");
  bot.pose("typing");
}
```

hd1370a - Source code



```
> fix the failing servo test

* Reading test/servos/test_park.cpp
* The test expects the elbow to park at
  0.0, but the default is now -0.1.
* Updating the expected value...
  - TEST_ASSERT_EQUAL_FLOAT(0.0f, pose);
  + TEST_ASSERT_EQUAL_FLOAT(-0.1f, pose);
* Ran pio test -e native: 42 passed

> _
```

hd1370a - AI agent chat

Tiny Engineer - laptop screen inserts: Paper screens

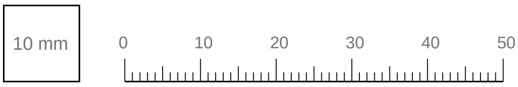
Servo preset: PowerHD HD-1370A (hd1370a)

Print only one page for hd1370a parts.

Any printer, any paper colour: no fill, so the paper becomes the screen.

Insert 32.7 x 25.2 mm (cut on the grey line). Visible window 32.3 x 16.7 mm, 0.2 mm from the left and 1.7 mm from the top edge.

Slide the paper insert, then a same-size piece of clear PET, into the lid pocket in front of LaptopScreen.



Square must measure 10 x 10 mm and the ruler 50 mm.

If not, reprint at 100% / Actual size (scaling off).

```
~/tiny-engineer$ pio run
Processing esp32-c3-devkitm-1 (espressif32)
------------------------------------------
Compiling .pio/build/src/anim/poses.cpp.o
Compiling .pio/build/src/http/routes.cpp.o
Compiling .pio/build/src/servos/park.cpp.o
Linking .pio/build/firmware.elf
Checking size .pio/build/firmware.elf
RAM:   [==        ]  17.4% (57012 bytes)
Flash: [=====     ]  52.1% (1023890 bytes)
Building .pio/build/firmware.bin
========== [SUCCESS] Took 12.84 seconds ==========
~/tiny-engineer$ _
```

hd1370a - Build output

```
// wave hello, then get back to work
void greet(Robot &bot) {
  for (int i = 0; i < 3; i++) {
    bot.arm(Side::Right, 0.8f);
    delay(180);
    bot.arm(Side::Right, -0.2f);
    delay(180);
  }
  bot.say("Hello, world!");
  bot.pose("typing");
}
```

hd1370a - Source code

```
> fix the failing servo test

* Reading test/servos/test_park.cpp
* The test expects the elbow to park at
  0.0, but the default is now -0.1.
* Updating the expected value...
  - TEST_ASSERT_EQUAL_FLOAT(0.0f, pose);
  + TEST_ASSERT_EQUAL_FLOAT(-0.1f, pose);
* Ran pio test -e native: 42 passed

> _
```

hd1370a - AI agent chat

```
Guru Meditation Error: Core 0 panic'ed
(Load access fault). Exception was unhandled.

Core 0 register dump:
MEPC    : 0x42008a3e  RA      : 0x42008a1c
SP      : 0x3fc9b2f0  GP      : 0x3fc8e400
MTVAL   : 0x00000000  <- null pointer

Backtrace:
  0x42008a3e in Coffee::refill() at mug.cpp:42
  0x42004f10 in Engineer::morning() at day.cpp:7
  0x42003c88 in loop() at main.cpp:118

Rebooting...
```

hd1370a - Crash / stack trace

```
**** TINY ENGINEER BASIC V2 ****
 64K RAM SYSTEM  38911 BASIC BYTES FREE

READY.
10 PRINT "HELLO, HUMAN"
20 GOTO 10
RUN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
BREAK IN 10
READY.
_
```

hd1370a - Retro computer

```
HOW DO YOU DO. PLEASE STATE YOUR PROBLEM.
> My robot keeps waving at me.
WHY DO YOU SAY YOUR ROBOT KEEPS WAVING
AT YOU?
> Maybe it wants a coffee.
YOU DON'T SEEM QUITE CERTAIN.
> I am never certain before coffee.
HOW LONG HAVE YOU BEEN NEVER CERTAIN
BEFORE COFFEE?
> Since I started writing firmware.
TELL ME MORE ABOUT WRITING FIRMWARE.
> _
```

hd1370a - ELIZA-style chat

```
 .-----.
 | o o |
 |  -  |
 '--+--'
/|   |\
 |___|

   TINY ENGINEER
booting desk robot...
[##########--------]
```

hd1370a - Tiny Engineer splash

```
draw or glue your screen inside
the dashed line (visible area)
```

hd1370a - Blank template

```
~/tiny-engineer$ pio run
Processing esp32-c3-devkitm-1 (espressif32)
------------------------------------------
Compiling .pio/build/src/anim/poses.cpp.o
Compiling .pio/build/src/http/routes.cpp.o
Compiling .pio/build/src/servos/park.cpp.o
Linking .pio/build/firmware.elf
Checking size .pio/build/firmware.elf
RAM:   [==        ]  17.4% (57012 bytes)
Flash: [=====     ]  52.1% (1023890 bytes)
Building .pio/build/firmware.bin
========== [SUCCESS] Took 12.84 seconds ==========
~/tiny-engineer$ _
```

hd1370a - Build output

```
// wave hello, then get back to work
void greet(Robot &bot) {
  for (int i = 0; i < 3; i++) {
    bot.arm(Side::Right, 0.8f);
    delay(180);
    bot.arm(Side::Right, -0.2f);
    delay(180);
  }
  bot.say("Hello, world!");
  bot.pose("typing");
}
```

hd1370a - Source code

```
> fix the failing servo test

* Reading test/servos/test_park.cpp
* The test expects the elbow to park at
  0.0, but the default is now -0.1.
* Updating the expected value...
  - TEST_ASSERT_EQUAL_FLOAT(0.0f, pose);
  + TEST_ASSERT_EQUAL_FLOAT(-0.1f, pose);
* Ran pio test -e native: 42 passed

> _
```

hd1370a - AI agent chat

```
Guru Meditation Error: Core 0 panic'ed
(Load access fault). Exception was unhandled.

Core 0 register dump:
MEPC    : 0x42008a3e  RA      : 0x42008a1c
SP      : 0x3fc9b2f0  GP      : 0x3fc8e400
MTVAL   : 0x00000000  <- null pointer

Backtrace:
  0x42008a3e in Coffee::refill() at mug.cpp:42
  0x42004f10 in Engineer::morning() at day.cpp:7
  0x42003c88 in loop() at main.cpp:118

Rebooting...
```

hd1370a - Crash / stack trace

```
**** TINY ENGINEER BASIC V2 ****
 64K RAM SYSTEM  38911 BASIC BYTES FREE

READY.
10 PRINT "HELLO, HUMAN"
20 GOTO 10
RUN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
BREAK IN 10
READY.
_
```

hd1370a - Retro computer

```
HOW DO YOU DO. PLEASE STATE YOUR PROBLEM.
> My robot keeps waving at me.
WHY DO YOU SAY YOUR ROBOT KEEPS WAVING
AT YOU?
> Maybe it wants a coffee.
YOU DON'T SEEM QUITE CERTAIN.
> I am never certain before coffee.
HOW LONG HAVE YOU BEEN NEVER CERTAIN
BEFORE COFFEE?
> Since I started writing firmware.
TELL ME MORE ABOUT WRITING FIRMWARE.
> _
```

hd1370a - ELIZA-style chat

```
 .-----.
 | o o |
 |  -  |
 '--+--'
/|   |\
 |___|

   TINY ENGINEER
booting desk robot...
[##########--------]
```

hd1370a - Tiny Engineer splash

```
draw or glue your screen inside
the dashed line (visible area)
```

hd1370a - Blank template

```
~/tiny-engineer$ pio run
Processing esp32-c3-devkitm-1 (espressif32)
------------------------------------------
Compiling .pio/build/src/anim/poses.cpp.o
Compiling .pio/build/src/http/routes.cpp.o
Compiling .pio/build/src/servos/park.cpp.o
Linking .pio/build/firmware.elf
Checking size .pio/build/firmware.elf
RAM:   [==        ]  17.4% (57012 bytes)
Flash: [=====     ]  52.1% (1023890 bytes)
Building .pio/build/firmware.bin
========== [SUCCESS] Took 12.84 seconds ==========
~/tiny-engineer$ _
```

hd1370a - Build output

```
// wave hello, then get back to work
void greet(Robot &bot) {
  for (int i = 0; i < 3; i++) {
    bot.arm(Side::Right, 0.8f);
    delay(180);
    bot.arm(Side::Right, -0.2f);
    delay(180);
  }
  bot.say("Hello, world!");
  bot.pose("typing");
}
```

hd1370a - Source code

```
> fix the failing servo test

* Reading test/servos/test_park.cpp
* The test expects the elbow to park at
  0.0, but the default is now -0.1.
* Updating the expected value...
  - TEST_ASSERT_EQUAL_FLOAT(0.0f, pose);
  + TEST_ASSERT_EQUAL_FLOAT(-0.1f, pose);
* Ran pio test -e native: 42 passed

> _
```

hd1370a - AI agent chat

```
Guru Meditation Error: Core 0 panic'ed
(Load access fault). Exception was unhandled.

Core 0 register dump:
MEPC    : 0x42008a3e  RA      : 0x42008a1c
SP      : 0x3fc9b2f0  GP      : 0x3fc8e400
MTVAL   : 0x00000000  <- null pointer

Backtrace:
  0x42008a3e in Coffee::refill() at mug.cpp:42
  0x42004f10 in Engineer::morning() at day.cpp:7
  0x42003c88 in loop() at main.cpp:118

Rebooting...
```

hd1370a - Crash / stack trace

```
**** TINY ENGINEER BASIC V2 ****
 64K RAM SYSTEM  38911 BASIC BYTES FREE

READY.
10 PRINT "HELLO, HUMAN"
20 GOTO 10
RUN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
BREAK IN 10
READY.
_
```

hd1370a - Retro computer

```
HOW DO YOU DO. PLEASE STATE YOUR PROBLEM.
> My robot keeps waving at me.
WHY DO YOU SAY YOUR ROBOT KEEPS WAVING
AT YOU?
> Maybe it wants a coffee.
YOU DON'T SEEM QUITE CERTAIN.
> I am never certain before coffee.
HOW LONG HAVE YOU BEEN NEVER CERTAIN
BEFORE COFFEE?
> Since I started writing firmware.
TELL ME MORE ABOUT WRITING FIRMWARE.
> _
```

hd1370a - ELIZA-style chat

```
 .-----.
 | o o |
 |  -  |
 '--+--'
/|   |\
 |___|

   TINY ENGINEER
booting desk robot...
[##########--------]
```

hd1370a - Tiny Engineer splash

```
draw or glue your screen inside
the dashed line (visible area)
```

hd1370a - Blank template

```
~/tiny-engineer$ pio run
Processing esp32-c3-devkitm-1 (espressif32)
------------------------------------------
Compiling .pio/build/src/anim/poses.cpp.o
Compiling .pio/build/src/http/routes.cpp.o
Compiling .pio/build/src/servos/park.cpp.o
Linking .pio/build/firmware.elf
Checking size .pio/build/firmware.elf
RAM:   [==        ]  17.4% (57012 bytes)
Flash: [=====     ]  52.1% (1023890 bytes)
Building .pio/build/firmware.bin
========== [SUCCESS] Took 12.84 seconds ==========
~/tiny-engineer$ _
```

hd1370a - Build output

```
// wave hello, then get back to work
void greet(Robot &bot) {
  for (int i = 0; i < 3; i++) {
    bot.arm(Side::Right, 0.8f);
    delay(180);
    bot.arm(Side::Right, -0.2f);
    delay(180);
  }
  bot.say("Hello, world!");
  bot.pose("typing");
}
```

hd1370a - Source code

```
> fix the failing servo test

* Reading test/servos/test_park.cpp
* The test expects the elbow to park at
  0.0, but the default is now -0.1.
* Updating the expected value...
  - TEST_ASSERT_EQUAL_FLOAT(0.0f, pose);
  + TEST_ASSERT_EQUAL_FLOAT(-0.1f, pose);
* Ran pio test -e native: 42 passed

> _
```

hd1370a - AI agent chat

```
Guru Meditation Error: Core 0 panic'ed
(Load access fault). Exception was unhandled.

Core 0 register dump:
MEPC    : 0x42008a3e  RA      : 0x42008a1c
SP      : 0x3fc9b2f0  GP      : 0x3fc8e400
MTVAL   : 0x00000000  <- null pointer

Backtrace:
  0x42008a3e in Coffee::refill() at mug.cpp:42
  0x42004f10 in Engineer::morning() at day.cpp:7
  0x42003c88 in loop() at main.cpp:118

Rebooting...
```

hd1370a - Crash / stack trace

```
**** TINY ENGINEER BASIC V2 ****
 64K RAM SYSTEM  38911 BASIC BYTES FREE

READY.
10 PRINT "HELLO, HUMAN"
20 GOTO 10
RUN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
HELLO, HUMAN
BREAK IN 10
READY.
_
```

hd1370a - Retro computer

```
HOW DO YOU DO. PLEASE STATE YOUR PROBLEM.
> My robot keeps waving at me.
WHY DO YOU SAY YOUR ROBOT KEEPS WAVING
AT YOU?
> Maybe it wants a coffee.
YOU DON'T SEEM QUITE CERTAIN.
> I am never certain before coffee.
HOW LONG HAVE YOU BEEN NEVER CERTAIN
BEFORE COFFEE?
> Since I started writing firmware.
TELL ME MORE ABOUT WRITING FIRMWARE.
> _
```

hd1370a - ELIZA-style chat

```
 .-----.
 | o o |
 |  -  |
 '--+--'
/|   |\
 |___|

   TINY ENGINEER
booting desk robot...
[##########--------]
```

hd1370a - Tiny Engineer splash

```
draw or glue your screen inside
the dashed line (visible area)
```

hd1370a - Blank template

```
~/tiny-engineer$ pio run
Processing esp32-c3-devkitm-1 (espressif32)
------------------------------------------
Compiling .pio/build/src/anim/poses.cpp.o
Compiling .pio/build/src/http/routes.cpp.o
Compiling .pio/build/src/servos/park.cpp.o
Linking .pio/build/firmware.elf
Checking size .pio/build/firmware.elf
RAM:   [==        ]  17.4% (57012 bytes)
Flash: [=====     ]  52.1% (1023890 bytes)
Building .pio/build/firmware.bin
========== [SUCCESS] Took 12.84 seconds ==========
~/tiny-engineer$ _
```

hd1370a - Build output

```
// wave hello, then get back to work
void greet(Robot &bot) {
  for (int i = 0; i < 3; i++) {
    bot.arm(Side::Right, 0.8f);
    delay(180);
    bot.arm(Side::Right, -0.2f);
    delay(180);
  }
  bot.say("Hello, world!");
  bot.pose("typing");
}
```

hd1370a - Source code

```
> fix the failing servo test

* Reading test/servos/test_park.cpp
* The test expects the elbow to park at
  0.0, but the default is now -0.1.
* Updating the expected value...
  - TEST_ASSERT_EQUAL_FLOAT(0.0f, pose);
  + TEST_ASSERT_EQUAL_FLOAT(-0.1f, pose);
* Ran pio test -e native: 42 passed

> _
```

hd1370a - AI agent chat